Off-centre CoE cosmology: the mathematical workflow from geometry to the Planck TT spectrum

Working notes for C. A. (revision 5, compiled 24 October 2026)

Abstract

We describe a toy universe in which a finite expansion of ordinary matter, evolving under classical laws, has a centre of expansion (CoE), and the observer sits off that centre inside a thick shell of matter, at $D = 58.2\,\mathrm{Gpc}$ in the working configuration. Because the symmetry is axial about the CoE axis rather than spherical about the observer, the model predicts a direction-dependent sky. We compare it with three data sets. (i) The cluster L_X–T normalisation A/A_{all} and slope B/B_{all} of Migkas et al.: with a density law locked to the CatWISE number-count dipole, the null $\chi^2 = 45.14$ over 71 longitude bins falls to about 10–12, and to 7.67 for the best direct fit; the predicted maximum and minimum bands are placed on the Migkas maps by a fixed tilt, gradient and roll. (ii) The CatWISE dipole, which fixes the local density gradient, $\rho'/\rho \approx -0.00106\,\mathrm{Gpc}^{-1}$. (iii) The Planck TT spectrum: the same geometry carried to last scattering gives a CMB far less uniform than observed, so the model needs matter structure around the observer that keeps the sky flat; with a stochastic field, a fixed low-ℓ background and a damped half-scale term the fit reaches $\chi^2_{2-2500} = 3{,}787$ and $r^2 = 0.976$, against 2,558 and 0.983 for ΛCDM. The A and B residuals left by the model, after shared harmonics $n = 1$–3, agree in sign with 11 of 12 independently known sky anomalies. Methodologically, the density law $m(r)$ is solved for numerically from the observations as an inverse problem, with each analytic law treated as a trial form. Goodness of fit is given only as χ^2 and Pearson r^2. Items not traced to code or data are marked [unverified: …].

1 Introduction

The standard cosmological model rests on the cosmological principle: on large scales the universe is homogeneous and isotropic, and every observer sees the same expansion. The inferred acceleration of that expansion, assigned to dark energy or Λ, comes from supernova distances that lie above the prediction of a decelerating homogeneous model. Several observations sit uneasily with exact isotropy. The X-ray scaling relations of galaxy clusters vary with direction (Migkas et al.), and the number-count dipole of CatWISE quasars is larger than the kinematic dipole expected from our motion relative to the CMB (Secrest et al. 2021). This paper asks whether an off-centre observer in an inhomogeneous but classically evolving expansion would see such anisotropies, and whether such an observer could read an apparent acceleration from distances without any acceleration taking place. The question is posed, not answered.

The toy universe is simple to state. Ordinary matter expands from a centre of expansion (CoE) under classical laws, with no dark matter and no cosmological constant. The observer is at distance D from the CoE, inside a thick shell of matter, with blast-wave-like radial structure: density and velocity depend on the distance r from the CoE. Lines of sight toward and away from the CoE cross different densities and recession velocities, so the sky seen by the observer has an axis. The model is confronted first with the cluster anisotropy, which it should reproduce from the off-centre geometry alone, and with the CatWISE dipole, which fixes its local density gradient. It must then face the CMB: carried to last scattering, the same geometry would make the CMB as anisotropic as the clusters, yet the observed CMB is isotropic to $\sim 10^{-5}$ after the dipole.

The method is that of an inverse problem in the sense used by Ellis and Stoeger for fitting cosmological models to observations: the density law is solved for from the data rather than assumed, and the residuals left by each step are examined for structure, following the tradition of Tukey (1977), Box (1976) and Mayo (1996). Section 2 sets out the model, §3 the numerical methods, §4 the data, §5 the results, §6 the tensions and limits, and §7 the conclusions. Symbols are listed in Table 1.

2 The model

2.1 The question and set-up

The residual here is the supernova excess itself. The standard route assumes a homogeneous model and fits a dark-energy term to what is left. The CoE toy model instead proposes a finite geometry of ordinary matter, under classical laws and seen from off its centre, that would leave the same residual without dark matter or dark energy.

The question. We separate *physical* expansion, the true radial motion of matter away from a centre of expansion, from *inferred* expansion, what an off-centre observer reconstructs from redshifts and luminosity distances under homogeneous–isotropic assumptions. The question is

Table 1: Parameters and symbols.

D, d	observer–CoE distance 58.2; vision radius 14.3005 Gpc
R, χ	source-sphere radius; distance from observer
$a_{\rm eq}$	equality scale factor, 1/3400
L, g	density scale length, shape
ϵ, s_R	web amplitude fraction; tidal sky at R
$\lg A$, n	source amplitude, tilt
A_w, r_s, ϕ	acoustic amplitude, scale, phase
k_D	damping cutoff
R_c, w	window centre, width
B, k_B, r_2, ϕ_2	damped mode
K_M	moving-lens kernel
c_ℓ	exact/Limber ratio
$D_\ell^{\rm fix}$	deterministic background, $\ell \le 30$
$\Delta\ell$, L	residual period, envelope

whether the second can look accelerated when the first is not. The geometry of Fig. 1 is the candidate; nothing in it is new physics.

Terms and symbols. **CoE**: centre of expansion, the point from which matter moves radially outward. D: distance from the observer to the CoE; working value 58.2 Gpc. d: the observer's vision (sampling) radius for the cluster data, 14.3005 Gpc, fixed at the Migkas sample depth and not fitted, so lines of sight sample radii $D - d \le r \le D + d$ from the CoE. R: radius of the CMB source sphere around the observer (comoving distance to last scattering in this model), explored at $R = 4$, 8, 14 Gpc. **Shell and core**: the observer sits inside a thick shell of ordinary matter around the CoE; the core is the matter nearer the CoE than the shell. $\rho(r)$: matter density as a function of distance r from the CoE, written $\rho_0 g(r/L)$ with scale length L and shape g (§5.1). V_E: the expansion (free-fall) velocity of matter relative to the CoE at radius r; at the observer's position it is the observer's own velocity $\vec{V}_O$, at a galaxy it is $\vec{V}_g$. $V_s(\psi)$: the recession velocity the observer measures along a line of sight at angle ψ, the local Hubble velocity $\vec{V}_H = \vec{V}_g - \vec{V}_O$ projected on that line. ψ: angle between a line of sight and the observer–CoE axis; $\psi = 0$ points away from the CoE (anti-CoE), $\psi = 180°$ toward it. l, b: Galactic longitude and latitude, the sky coordinates in which the data are given; the CoE axis points to longitude l_0, so ψ is a function of (l, b). A, B: normalisation and slope of the cluster L_X–T scaling relation, $L_X \propto A\,T^B$; $A/A_{\rm all}$ and $B/B_{\rm all}$ are their values in a sky cone relative to the all-sky fit. Symmetry is axial about the CoE axis, not spherical about the observer, so the sky anisotropy is an off-centre sampling of radial structure, not a direction-dependent scale factor.

How the model is tested. The standard route assumes FLRW or ΛCDM, predicts $D_L(z)$ and fits H_0, Ω_m, w. Here the CoE geometry is fixed first and then asked what it predicts. The first observables it must match are

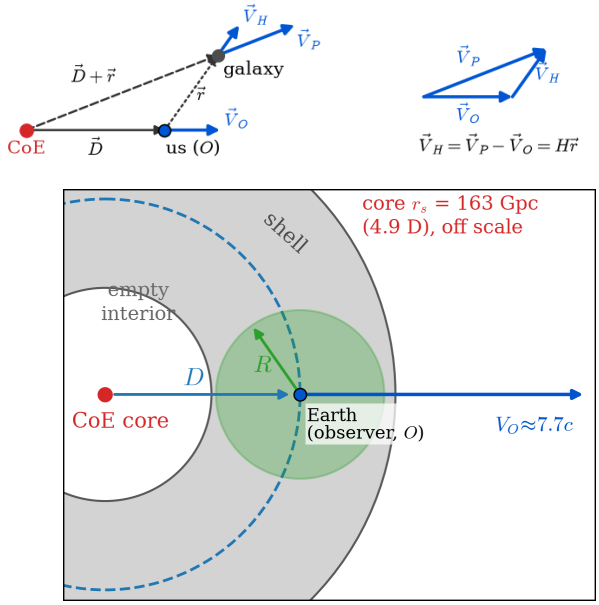


Figure 1: Set-up. Top: Hubble velocity vectors for the CoE, the observer and a galaxy, $\vec{V}_H = \vec{V}_g - \vec{V}_O$. Bottom: the observer inside a thick ordinary-matter shell at distance D from the CoE, with its vision sphere. Restored from the original preprint (Figs. 1–2 there).

A and B against angle (§4.1); the second is the CMB (§5.3).

2.2 Geometry and time dependence

The reference model is a homogeneous expansion seen from its centre. This section sets out the geometry for an observer at distance D from the centre, which is what any residual relative to the homogeneous model has to be computed from.

Set-up and symbols. D: distance from the observer to the centre of expansion (CoE); working value $D = 58.2$ Gpc. R: the observer's vision radius, i.e. the comoving distance to the CMB source surface, explored at $R = 4$, 8, 14 Gpc. Outer edge of the matter: $1.25D$ (the figure uses 72.83 Gpc). No cosmological constant; dynamics are matter plus early radiation.

Volumes (worked). A sphere of radius r has $V = \frac{4}{3}\pi r^3$. Observer sphere, $R = 4$: $V_R = 4.18879 \times 64 = 268.1\,\mathrm{Gpc}^3$. Sphere to $D + R = 62.2$ Gpc: $4.18879 \times 62.2^3 = 1.008 \times 10^6\,\mathrm{Gpc}^3$, so $V_R/V_{D+R} = 268.1/1.008 \times 10^6 = 0.0266\%$. Outer sphere 72.83 Gpc: $1.618 \times 10^6\,\mathrm{Gpc}^3$, so $V_R/V_{\rm out} = 0.0166\%$. (Arithmetic checked here; the 72.83 vs $1.25 \times 58.2 = 72.75$ difference is in the source figure.)

Radiation and matter densities. Inputs: $a(t)$, the scale factor ($a = 1$ today), and $a_{\rm eq} = 1/3400$. *Step 1*: matter is conserved particles in a volume $\propto a^3$, so $\rho_m = \rho_{m,0} a^{-3}$. *Step 2*: photons are also diluted as a^{-3}, but each

photon's energy $h\nu \propto a^{-1}$ (redshift), so $\rho_r \propto a^{-4}$. *Step 3*: write $\rho_r = \rho_{r,0}a^{-4}$ and define equality by $\rho_r(a_{\rm eq}) = \rho_m(a_{\rm eq})$. Then $\rho_{r,0} = a_{\rm eq}\rho_{m,0}$, so

$$\frac{\rho_r}{\rho_{m,0}} = \frac{a_{\rm eq}}{a^4}, \qquad \frac{\rho_m}{\rho_{m,0}} = \frac{1}{a^3}. \tag{1}$$

Example: at $a = 10^{-4}$, $\rho_r/\rho_{m,0} = 2.94 \times 10^{12}$ and $\rho_m/\rho_{m,0} = 10^{12}$, so radiation dominates by a factor of $a_{\rm eq}/a = 2.94$.

From densities to $a(t)$. The Friedmann equation without Λ or curvature is $H^2 \equiv (\dot{a}/a)^2 = \frac{8\pi G}{3}(\rho_r + \rho_m)$. Substituting Eq. (1) gives $H = H_*\sqrt{a_{\rm eq}a^{-4} + a^{-3}} = H_* a^{-2}\sqrt{a_{\rm eq} + a}$ with $H_*^2 = 8\pi G\rho_{m,0}/3$. Since $dt = da/(aH)$,

$$H_* t(a) = \int_0^a \frac{a'\,da'}{\sqrt{a_{\rm eq} + a'}} = \tfrac{2}{3}(a - 2a_{\rm eq})\sqrt{a + a_{\rm eq}} + \tfrac{4}{3}a_{\rm eq}^{3/2}. \tag{2}$$

The intermediate step is the substitution $u = a' + a_{\rm eq}$, giving $\int (u - a_{\rm eq})u^{-1/2}du$. H_* is chosen so that $t(1) = 13.8\,$Gyr; this is an assumption, not an output. Plugging in: $t(a_{\rm eq}) = 40.8\,$kyr and $t(1/1090) = 297.5\,$kyr (recombination). Limits: for $a \ll a_{\rm eq}$, $H_* t \approx a^2/(2\sqrt{a_{\rm eq}})$, so $a \propto t^{1/2}$; for $a \gg a_{\rm eq}$, $H_* t \approx \frac{2}{3}a^{3/2}$, so $a \propto t^{2/3}$.
Validity. Standard homogeneous FRW dilution. It is put in by assumption for the CoE universe. It would fail if the CoE dynamics are not FRW-like (inhomogeneous LTB shells evolve at different rates) or if $D(t)$ from the Newtonian integrator differs; that replacement is listed as future work.

Volume growth rates. If a length $\propto t^p$, then $V \propto t^{3p} \equiv t^n$ and $\dot{V}/V = n/t$. Radiation era $p = \frac{1}{2}$, so $n = \frac{3}{2}$. For D in the matter era $p = \frac{2}{3}$, so $n = 2$; today $2/13.8\,{\rm Gyr} = 0.145\,{\rm Gyr}^{-1}$. For the horizon $R \propto t$, so $n = 3$.

3 Methods

3.1 The inverse approach

The density law is not known in advance. We treat it as the unknown of an inverse problem: the off-centre geometry is fixed, Eq. (8) is the forward map from $m(r)$ to the sky, and $m(r)$ is solved for numerically from the observations. The data do not fix $m(r)$ uniquely, so each analytic law in §5.1 is a trial form; the CatWISE lock removes one degree of freedom from each. After each fit the residual $R = Y - M$ is inspected. Where it shows structure, a further term is proposed, fitted and tested on the next observable. The same approach is used for the CMB, where the unknown is the distribution of matter around the observer that produces a nearly flat sky and the Planck spectrum. Because C_ℓ discards the phases of the $a_{\ell m}$, that problem is also degenerate: different arrangements of over- and under-densities produce the same target (§3.3).

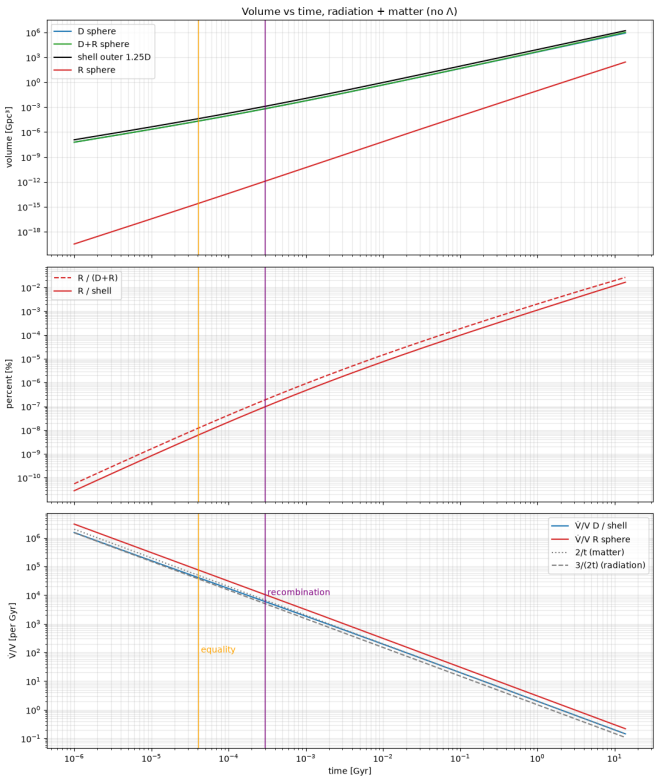


Figure 2: Volume versus time with radiation + matter (no Λ). Top: volumes of the D sphere, $D{+}R$ sphere and outer sphere from $a(t)$ of Eq. (2). Middle: V_R as a percentage of the larger volumes. Bottom: $\dot{V}/V = n/t$. Vertical lines: equality (41 kyr), recombination (297 kyr).

3.2 Fitting the cluster profiles

For the 71 longitude bins, $\chi^2 = \sum_{i=1}^{71}(y_i - m_i)^2/\sigma_i^2$ with $y_i = A/A_{\rm all}$ and Migkas errors σ_i; the null model is $m_i = 1$. For each locked law, D and the axis longitude l_0 are refitted while L stays locked. The model is then placed on the full sky in three fixed steps (tilt, radial gradient, roll about the CoE axis) and its predicted maximum and minimum bands are compared with the Migkas maps; this adds no morphological degree of freedom.

3.3 Spherical harmonics, C_ℓ, D_ℓ and cosmic variance

Spherical harmonics are the bookkeeping for sky residuals: they separate a map into scales, and cosmic variance sets how large a residual at each ℓ has to be before it means anything.

Expansion. Input: a sky map $\Theta(\hat{n}) \equiv \Delta T/T$ (dimensionless). Since the $Y_{\ell m}$ are orthonormal on the sphere,

$$\Theta(\hat{n}) = \sum_{\ell m} a_{\ell m}Y_{\ell m}(\hat{n}), \quad a_{\ell m} = \int d\Omega\,\Theta Y^*_{\ell m}. \tag{3}$$

ℓ is the number of nodal lines; a mode of degree ℓ has angular half-wavelength $\theta \approx 180^\circ/\ell$, so $\ell = 220$ gives 0.82°.

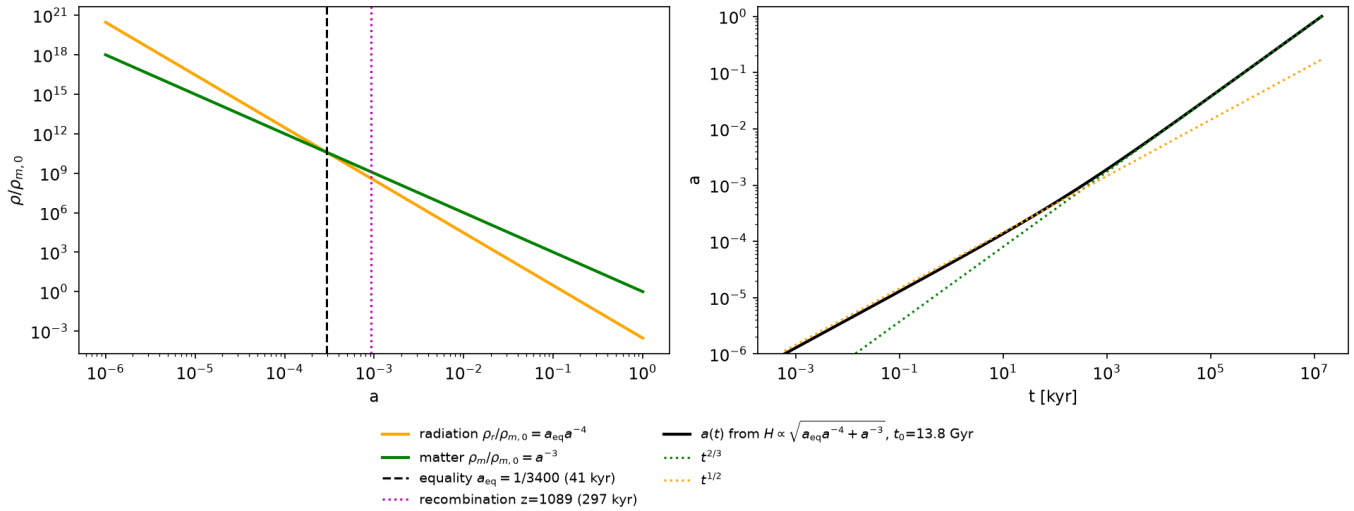


Figure 3: Eq. (1) (left, units of today's matter density) and Eq. (2) (right) for $a_{\rm eq} = 1/3400$, $t_0 = 13.8\,$Gyr. Dotted: the limits $t^{1/2}$ and $t^{2/3}$.

For a statistically isotropic sky, $\langle a_{\ell m} a^*_{\ell' m'} \rangle = C_\ell \delta_{\ell\ell'} \delta_{mm'}$, and the estimator from one sky is $\hat{C}_\ell = \frac{1}{2\ell+1} \sum_m |a_{\ell m}|^2$.

From C_ℓ to D_ℓ. The map variance is $\langle \Theta^2 \rangle = \sum_\ell \frac{2\ell+1}{4\pi} C_\ell$. Converting the sum to an integral over $\ln \ell$ gives $\sum_\ell \frac{2\ell+1}{4\pi} C_\ell \approx \int d\ln\ell \, \frac{\ell(2\ell+1)}{4\pi} C_\ell \approx \int d\ln\ell \, \frac{\ell(\ell+1)}{2\pi} C_\ell$. So

$$D_\ell \equiv \frac{\ell(\ell+1)}{2\pi} C_\ell \, T_0^2 \quad [\mu\mathrm{K}^2] \tag{4}$$

is the temperature variance per unit $\ln \ell$, with $T_0 = 2.7255 \times 10^6\,\mu$K. A scale-invariant Sachs–Wolfe plateau has $C_\ell \propto 1/[\ell(\ell+1)]$, so its D_ℓ is flat, which is why the plots use D_ℓ. Example: $C_\ell = 10^{-10}$ at $\ell = 10$ gives $D_\ell = 17.5 \times 10^{-10} \times 7.43 \times 10^{12} = 1.30 \times 10^4\,\mu\mathrm{K}^2$. In code, D_ℓ is computed from the map with `healpy.anafast` after removing the monopole and dipole. The dipole is removed because the observer's motion dominates it.

Cosmic variance. If the $a_{\ell m}$ are Gaussian, $(2\ell+1)\hat{C}_\ell/C_\ell$ is χ^2-distributed with $2\ell+1$ degrees of freedom (the m modes are real combinations). That distribution has variance $2(2\ell+1)$, so

$$\frac{\Delta C_\ell}{C_\ell} = \sqrt{\frac{2}{2\ell+1}}. \tag{5}$$

Values: 63% at $\ell = 2$, 31% at 10, 18% at 30, 6.7% at 220, 3.2% at 1000. This is the irreducible scatter of one sky about its ensemble mean, and it is the $\pm 1\sigma$ band drawn around every CoE curve.

Validity. It is exact for a Gaussian, isotropic, full-sky field. It underestimates the scatter for non-Gaussian skies, e.g. a few dominant voids or the axisymmetric CoE background, and for masked skies. It does not apply to a deterministic component, which has no ensemble scatter at all.

3.4 Goodness of fit: χ^2 and Pearson r^2

χ^2 is the weighted sum of squared residuals and Pearson r^2 measures how much shape they share with the data; neither by itself says what structure a residual has, which is why the residuals are also plotted.

χ^2. Inputs: data d_ℓ, model m_ℓ, Planck errors. Planck low-ℓ errors are asymmetric $(\sigma_\ell^-, \sigma_\ell^+)$, so the code uses the error on the side where the model lies:

$$\chi^2 = \sum_\ell \frac{(m_\ell - d_\ell)^2}{\sigma_\ell^2}, \quad \sigma_\ell = \begin{cases} \sigma_\ell^+ & m_\ell > d_\ell \\ \sigma_\ell^- & m_\ell \le d_\ell \end{cases}. \tag{6}$$

Meaning: each term is the squared distance in units of measurement error. For N independent Gaussian points and a correct model, $\langle \chi^2 \rangle \approx N - p$, with p the number of fitted parameters. Example: ΛCDM over ℓ 2–30 (29 points) gives 20.1. Here χ^2 uses Planck errors only, because Planck's errors already include cosmic variance for ΛCDM. The stochastic-field fits also used $\chi^2_{\rm CV}$, which adds the model's own cosmic variance $\frac{2}{2\ell+1} D^2_{\rm field}$ in quadrature.
Validity. It assumes independent Gaussian errors. It fails at low ℓ, where the likelihood is skewed (hence the asymmetric errors), and for the binned $\ell \ge 30$ file, where bins are not fully independent. Read χ^2 as a comparison measure, not an exact probability.

Pearson r^2. With means $\bar{d}$, $\bar{m}$,

$$r^2 = \frac{\left[\sum (d_\ell - \bar{d})(m_\ell - \bar{m})\right]^2}{\sum (d_\ell - \bar{d})^2 \sum (m_\ell - \bar{m})^2}. \tag{7}$$

It is the squared linear correlation and is invariant under $m \to am + b$. It measures shape agreement and ignores amplitude and offset; it is *not* a fit R^2. Example: ΛCDM at ℓ 2–30 has $r^2 = 0.012$ although $\chi^2 = 20.1$, because the data there are scatter around a nearly flat curve. Over ℓ

2–2500, where the peaks dominate the variance, ΛCDM has $r^2 = 0.983$. So r^2 is informative only where the signal varies a lot compared with the noise.

4 Data

4.1 Cluster scaling relations: $A/A_{\rm all}$ and $B/B_{\rm all}$ versus angle

For an isotropic universe both ratios would be 1 in every direction. What the data show is the residual from 1, and that residual, not a fitted curve, is what the CoE model must reproduce.

Migkas et al. test sky isotropy with the X-ray luminosity–temperature relation of galaxy clusters. L_X depends on luminosity distance and T does not, so a directional change in the normalisation A traces a directional distance or expansion scale. With $A\,H_0^2$ held fixed, $H/H_{\rm ref} = 1/\sqrt{A/A_{\rm all}}$, so troughs in $A/A_{\rm all}$ are directions of higher apparent expansion rate, provided the variation is cosmological rather than a cluster or X-ray systematic. The slope B is set more by sample composition. Both are given in sliding cones of width $\Delta l = 90°$ against Galactic longitude (their Fig. 6), digitised here at 71 points (Fig. 4).

Two features matter for what follows. $A/A_{\rm all}$ has two humps and two troughs over 360°, with its deepest low near $l \approx 255$–270°; $B/B_{\rm all}$ varies more slowly, with one broad low and one broad high. The curves in Fig. 4 are the author's spreadsheet model with a one-cycle phase rotation, $(1 \mp \mathrm{Amp}\,\sin(l+\varphi))$ times the shell series, Amp $= \mp 0.06$, $\varphi = 0.35$ rad; it is a diagnostic phase alignment, not a free dipole fit. Against the spreadsheet series the rotation raises R^2 for A from 0.501 to 0.739 and for B from 0.372 to 0.733; against the digitised points, from 0.446 to 0.712 (A) and from 0.483 to 0.785 (B). Source: `migkas_bars_rotation_summary.json`.

4.2 CatWISE, Planck and sky anomalies

CatWISE dipole (Secrest et al. 2021). The quasar number-count dipole $\mathcal{D} = 0.01554$, used only through the lock of Eq. (9).

Planck 2018 TT. The temperature power spectrum D_ℓ for $\ell = 2$–2500, with asymmetric errors at low ℓ and a binned file for $\ell \geq 30$. The ΛCDM curve is the Planck best fit.

Sky anomalies. Published positions (l, b) of the CMB dipole poles, bulk flow, CMB Cold Spot, NVSS and CatWISE dipoles, hemispherical power asymmetry and Loop I/North Polar Spur. Regions already used in the Migkas fit are excluded.

5 Results

5.1 Density laws, the CatWISE lock and the Migkas fit

Relative to an isotropic sky, the Migkas normalisation $A/A_{\rm all}$ has a residual dependence on direction. Here that residual is the observable, and the CoE density law is the model fitted to it. Seen as an inverse problem, which is how Ellis and Stoeger framed the fitting of cosmological models to observations, the density laws below are trial forms: the data do not fix $m(r)$ uniquely, and each law is one candidate that the off-centre geometry can carry; what the fit leaves is carried to the A/B decomposition in §5.10.

Observable. Migkas et al. measure the galaxy-cluster scaling-relation normalisation A in sky regions relative to its all-sky value, $y_i = A/A_{\rm all}$, in 71 bins of galactic longitude l with errors σ_i. In the CoE picture, a line of sight at angle ψ to the CoE axis crosses a different matter density and recession velocity from one at $\pi - \psi$, which produces a dipole-like modulation.

Line-of-sight geometry. Put the CoE at the origin and the observer at $\mathbf{D}$. A point at distance s from the observer in direction $\hat{n}$ has $r(s,\hat{n}) = |\mathbf{D} + s\hat{n}| = \sqrt{D^2 + s^2 + 2Ds\cos\psi}$, where $\cos\psi = \hat{D}\cdot\hat{n}$. Averaging a response f over the cluster depth distribution $p(s)$,

$$\frac{A}{A_{\rm all}}(\hat{n}) = \frac{\int ds\, p(s)\, f[\rho(r(s,\hat{n}))]}{\left\langle \int ds\, p(s)\, f[\rho]\right\rangle_{\rm sky}}. \tag{8}$$

The denominator normalises the all-sky mean to 1. Because the scale lengths below are $L \gg s$, expanding to first order gives $\rho(r) \approx \rho(D)[1 + (\rho'/\rho)_D\, s\cos\psi]$: a pure dipole in $\cos\psi$ with amplitude set by the local logarithmic gradient. [unverified: the exact form of f and $p(s)$ in the Migkas scripts was not re-derived here].

CatWISE lock (worked derivation). Inputs: shape $g(x)$, $x = r/L$; $D = 58.2$ Gpc; vision radius $d = 14.3005$ Gpc; $\Omega_m = 0.3$, $H_0 = 70$. *Step 1, normalisation*: $\rho(r) = \rho_0 g(r/L)$ with $\rho(D) = 0.3\rho_{\rm crit}$, so $\rho_0 = 0.3\rho_{\rm crit}/g(D/L)$. *Step 2, contrast*: require

$$\delta \equiv \frac{\rho(D-d) - \rho(D+d)}{\rho(D)} = 0.031. \tag{9}$$

This is twice the CatWISE number-count dipole $\mathcal{D} = 0.01554$ (Secrest et al. 2021; $2\mathcal{D} = 0.03108$, rounded). The factor 2 converts a dipole amplitude (peak-to-mean) into the peak-to-peak difference. *Step 3, exponential*: $g = e^{-x}$ gives $\rho(D \mp d)/\rho(D) = e^{\pm d/L}$, so $\delta = e^{d/L} - e^{-d/L} = 2\sinh(d/L)$ and

$$L = \frac{d}{\operatorname{arcsinh}(\delta/2)} = \frac{14.3005}{0.0154994} = 922.65 \text{ Gpc}. \tag{10}$$

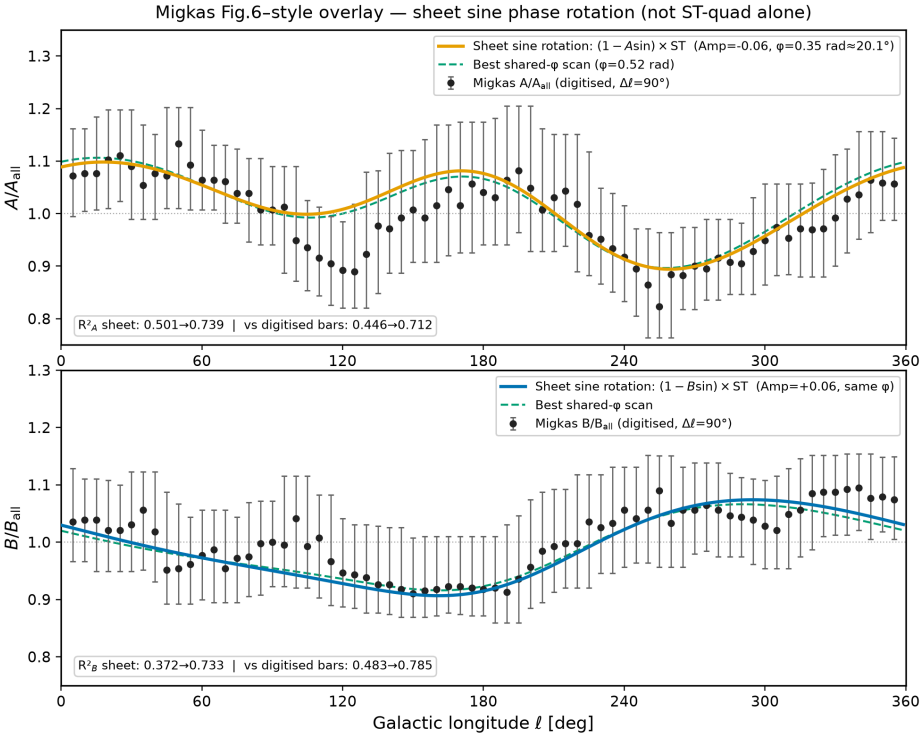


Figure 4: The observables. Top: Migkas $A/A_{\rm all}$ against Galactic longitude (digitised, $\Delta l = 90^\circ$ cones, black with error bars). Bottom: $B/B_{\rm all}$. Solid: spreadsheet shell model with the one-cycle sine rotation (Amp $= \mp 0.06$, $\varphi = 0.35$ rad); dashed: best shared-φ scan. R^2 before and after rotation is given in each panel. Restored from `cosmology/figures`.

Then $L/D = 15.85$, $\rho_0/\rho_{\rm crit} = 0.3\,e^{58.2/922.65} = 0.3195$, and the local gradient is $\rho'/\rho|_D = -1/L = -0.001084\,{\rm Gpc}^{-1}$. With $d = 3\,$Gpc instead, $L = 193.56\,$Gpc. *Step 4, other shapes*: Eq. (9) is solved numerically for L with the shape parameters held fixed (table). Every law reproduces $\delta = 0.031$ to 10^{-15}.

law	$g(x)$	L [Gpc]	$\rho_0/\rho_{\rm c}$
exp	e^{-x}	922.6	0.3195
power	$(1+x)^{-n}$, n=1.20	1049.3	0.3201
cored	$(1+x^2)^{-1}$	321.9	0.3098
Gauss	$e^{-x^2/2}$	231.5	0.3096
soft	$(1+x^\alpha)^{-\beta}$, α=0.30, β=1	5109.6	0.3784
stretched	e^{-x^η}, η=0.25	15246	0.3847
η-edge	e^{-x^η}, η=0.0500	0.849	1.0319

Meaning: across $D \pm d$ all laws have $\rho'/\rho \approx -0.00106$ to $-0.00109\,{\rm Gpc}^{-1}$. The lock fixes the local gradient, which is what the dipole measures, so inside the window the laws are indistinguishable (Fig. 5). They differ only near the CoE, where no data constrain them. The level-set scan imposed $\eta \geq 0.25$; the η-edge solution comes from a separate free-η fit. Sources: `catwise-density-levelset-fit.json`, `catwise-eta-migkas-fit.json`.

Validity. The first-order expansion needs $d \ll L$. For exp, $d/L = 0.0155$, so it holds. For the η-edge, $L = 0.849\,{\rm Gpc} \ll d$, so the law is far from linear across the window, even though δ is still met exactly.

Fit. $\chi^2 = \sum_{i=1}^{71}(y_i - m_i)^2/\sigma_i^2$ (§3.4). The null model $m_i = 1$ gives $\chi^2 = 45.14$. At fixed axis longitude $l_0 = 15.2^\circ$, with D refitted (so not 58.2) and L locked: exp 11.61 ($D = 48.26$), power 11.58 ($D = 47.86$), soft 10.43 ($D = 39.11$), stretched η=0.25 10.39 ($D = 38.47$). With l_0 free ($\approx 21.7^\circ$): cored 11.60 ($D = 59.5$), Gauss 11.61 ($D = 59.8$). Free η-edge: 9.28 ($D = 34.96$, $l_0 = 19.9^\circ$). The direct fits of Fig. 17 give 10.79 ($l_0 = 100^\circ$) and 7.67 ($l_0 = 15.6^\circ$), the latter equal to a reference sinusoid.

5.2 Rings, the Migkas overlay and rotation

The fit above matches A along longitude only. The residual it leaves is in the morphology on the full sky: where the maxima, minima and rings fall. That is tested here, with the geometry held fixed.

Rings. Off-centre placement gives closed curves of constant phase $\varphi(\psi)$ on the sky: rings about the CoE axis. In the original preprint the low-redshift ring and the longitude dependence of $A/A_{\rm all}$ were fixed from the geometry before the comparison with the Migkas maps (its Fix Point 1); the Migkas data were not used to add a morphological degree of freedom.

Tilt, gradient and roll. The full longitude profile, with the band positions and the residual bars they leave,

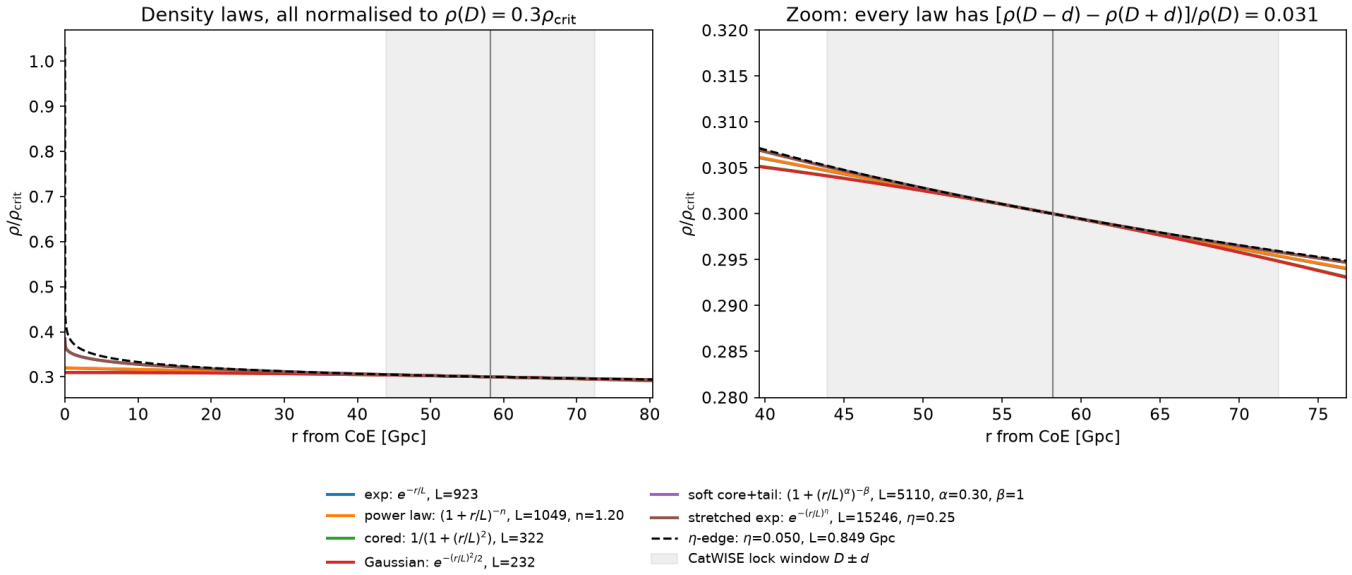


Figure 5: CatWISE-locked laws $\rho/\rho_{\rm crit}$ versus distance from the CoE (left) and zoom on $D \pm d$ (right), plotted from `catwise-density-levelset.csv` and `catwise-eta-density.csv`. $D = 58.2$ Gpc (line), $d = 14.3005$ Gpc (band).

is shown in Fig. 6. The model is placed on the sky in three steps, starting from an untilted configuration with no roll: a +15° tilt of the CoE axis (anti-CoE near $(l, b) = (0°, +15°)$), a radial gradient, and a +80.8° roll (yaw) about the CoE axis (Fig. 7). Fig. 8 overlays the resulting maximum and minimum bands on Migkas Fig. 8. Each band is 18° end to end (±9°, 5% of 360°). The Migkas maxima include $(17°, -9°)$ and $(185°, 0°)$. The residual maps in §5.10 use a later orientation of the same kind, a 15° lift and a 45° counterclockwise rotation; the two orientations are reported as they were made and have not been reconciled.

How much the rotation adds. In the original preprint's fits to $A/A_{\rm all}$ (null $\chi^2 = 45.14$), a point mass at the CoE without rotation gives $R^2 = 0.521$; with rotation and the one-cycle phase factor R^2 rises to 0.845 (exponential law, CatWISE lock) and 0.870 (soft core). These are unweighted R^2 values from that draft and are not recomputed here. The amplitude is partly set by how the construction is normalised (the density–distance trade); the morphology is the part that was not tuned.

5.3 The CMB: what keeps the sky flat

If the off-centre geometry is real, the anisotropy it puts into A and B should also appear on the last-scattering surface. The observed CMB is isotropic to $\sim 10^{-5}$ after the dipole. That near-absence is the residual the rest of these notes work on.

Holding the geometry fixed and carrying it to last scattering ($z_\star \approx 1100$), the original preprint found a geometric multipole of several percent on the CMB sky, orders of magnitude above the Planck isotropy level. Geometry alone is therefore incomplete. That draft then added special-relativistic Doppler and central-mass gravitational redshift, which together cancelled 95.9% of the geometric RMS while keeping the low-z morphology (its Fix Point 2). The model must therefore say what distribution of matter around the off-centre observer gives a nearly flat CMB sky and, at the same time, the Planck TT spectrum. The following subsections use the tools of §3 (harmonics, cosmic variance, χ^2) and build a realistic void/cluster web, the stochastic field and its projection, and then fit the spectrum as a fixed background plus acoustic and damped terms. The residuals that remain are taken back to the A/B sky in §5.10.

5.4 Matching target spectra

Primers. An axisymmetric pattern ($m = 0$) is a sum of Legendre polynomials $P_\ell(\cos\theta)$, so a single P_ℓ puts all its power in one ℓ (Fig. 9). N equal masses equally spaced on a ring have azimuthal Fourier power only at multiples of N (aliasing). Binary mixtures and quadrature masses behave the same way. *Matching a target*: given a target C_ℓ, find over-/under-densities that produce it. Because C_ℓ discards the phases of the $a_{\ell m}$, the solution is degenerate (Figs. 9–12).

Web and kernel ladder. Before targets were matched, hand-placed masses were replaced by a statistically realistic web of clusters and voids (Fig. 10), and its sky imprint was evaluated on spheres of radius R around the observer, the kernel ladder of Fig. 11. How the web is built and how the ladder is read are given in §5.5.

5.5 A realistic web of voids and clusters

Hand-placed masses are a model whose residuals depend on the hand. A statistical web replaces them so that what-

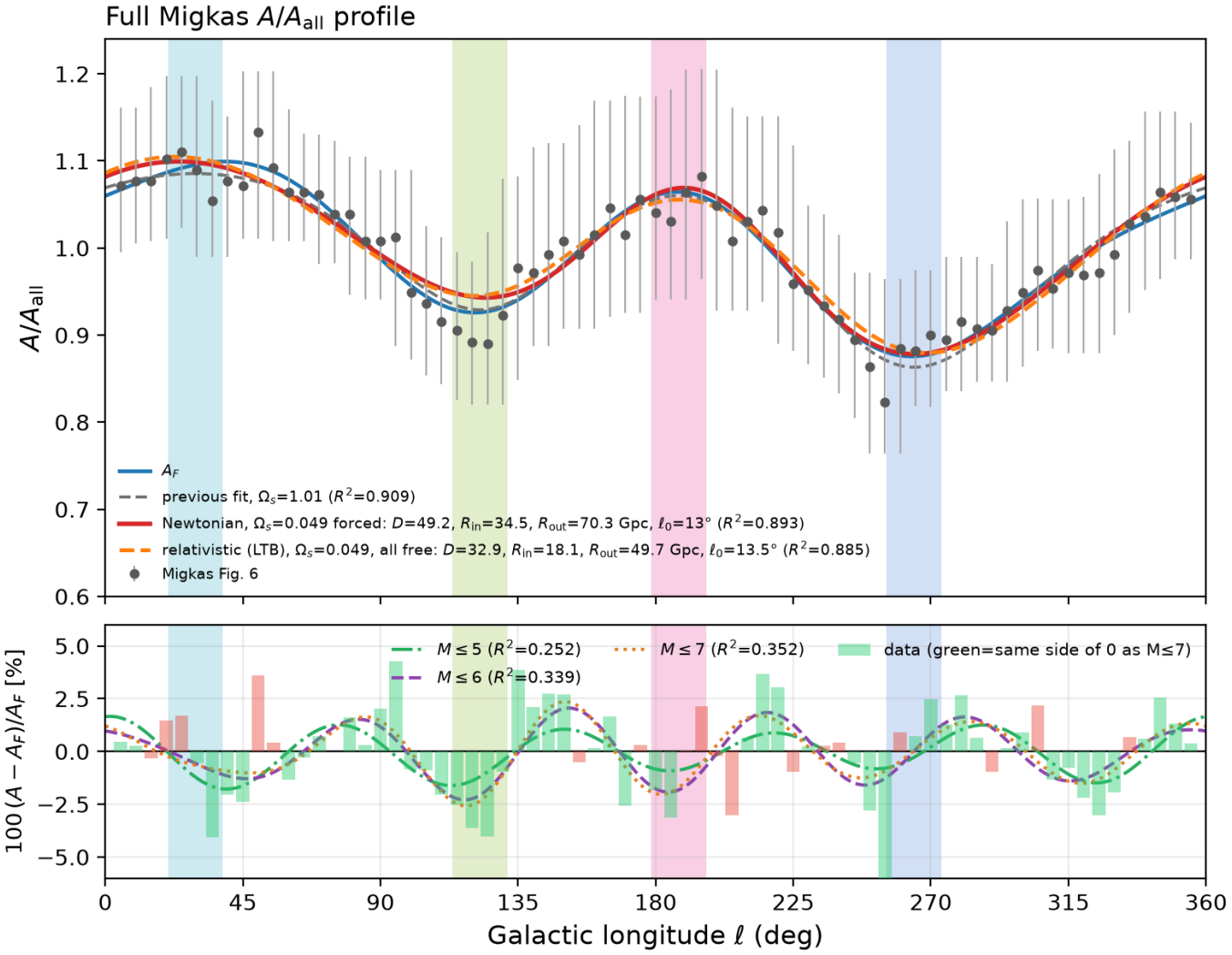


Figure 6: Full Migkas $A/A_{\rm all}$ profile against Galactic longitude with the model curves; coloured vertical bands mark the predicted MAX/MIN longitude bands. Bottom: residual bars, $100(A-A_F)/A_F$, with low-order harmonic fits ($M \le 5, 6, 7$). Pre-made figure from the original preprint.

ever structure remains in the projected field is not an artefact of where objects were put.

Purpose. Hand-placed masses are replaced by a statistically realistic population. Each body sources a Newtonian potential, and the sky signal is the gravitational field sampled on spheres around the observer.

Ingredients (from `realweb/build.py`, `cosmo.py`). Background: $h = 0.7$, $\Omega_m = 0.31$, $\Omega_b = 0.049$, $n_s = 0.965$, $\sigma_8 = 0.8$, $\bar\rho_m = \Omega_m\, 2.775\times10^{11}h^2 = 4.215\times10^{10}\,M_\odot{\rm Mpc}^{-3}$. Linear spectrum $P(k) = Ak^{n_s}T^2_{\rm EH}(k)$ (Eisenstein–Hu no-wiggle transfer function). A is fixed by $\sigma^2(R) = \int \frac{k^2dk}{2\pi^2}P(k)W^2(kR) = 0.8^2$ at $R = 8/h\,$Mpc, with the top-hat $W(x) = 3(\sin x - x\cos x)/x^3$.

Clusters, Press–Schechter. A region of mass $M = \frac{4}{3}\pi R^3\bar\rho_m$ collapses when its linear overdensity exceeds $\delta_c = 1.686$. The fraction above threshold is $\mathrm{erfc}(\nu/\sqrt{2})$ with $\nu = \delta_c/\sigma(M)$; differentiating with respect to $\ln M$ and multiplying by $\bar\rho_m/M$ gives

$$\frac{dn}{d\ln M} = \sqrt{\tfrac{2}{\pi}}\frac{\bar\rho_m}{M}\nu\left|\frac{d\ln\sigma}{d\ln M}\right|e^{-\nu^2/2}, \qquad (11)$$

for $M = 10^{14}$–$2\times10^{15}M_\odot$ (41 bins). The peak-background bias is $b = 1 + (\nu^2-1)/\delta_c$: halos are more numerous where the large-scale $\delta_{\rm lin} > 0$.

Voids, Sheth–van de Weygaert (2004). This is the two-barrier excursion-set size function with void threshold $\delta_v = -2.7$, evaluated at linear radius $R_L = R/1.7$ for $R = 10$–$50\,$Mpc. Bias -1.5. A void of radius R_v has interior contrast -0.85, i.e. mass deficit $M_d = 0.85\bar\rho_m\frac{4\pi}{3}R_v^3$. Example: $R_v = 30\,$Mpc gives $M_d = 4.0\times10^{15}M_\odot$. It is painted as $-M_d$ at the centre plus 8 random points of $+M_d/8$ on a shell at $1.25R_v$, so each void is compensated (net mass zero).

Placement. Counts per cell are Poisson with mean

$$\lambda = \frac{dn}{d\ln X}\Delta\ln X\,\Delta V\,\max(0, 1+b\,\delta_{\rm lin})\,m(\mathbf{x})\,f, \qquad (12)$$

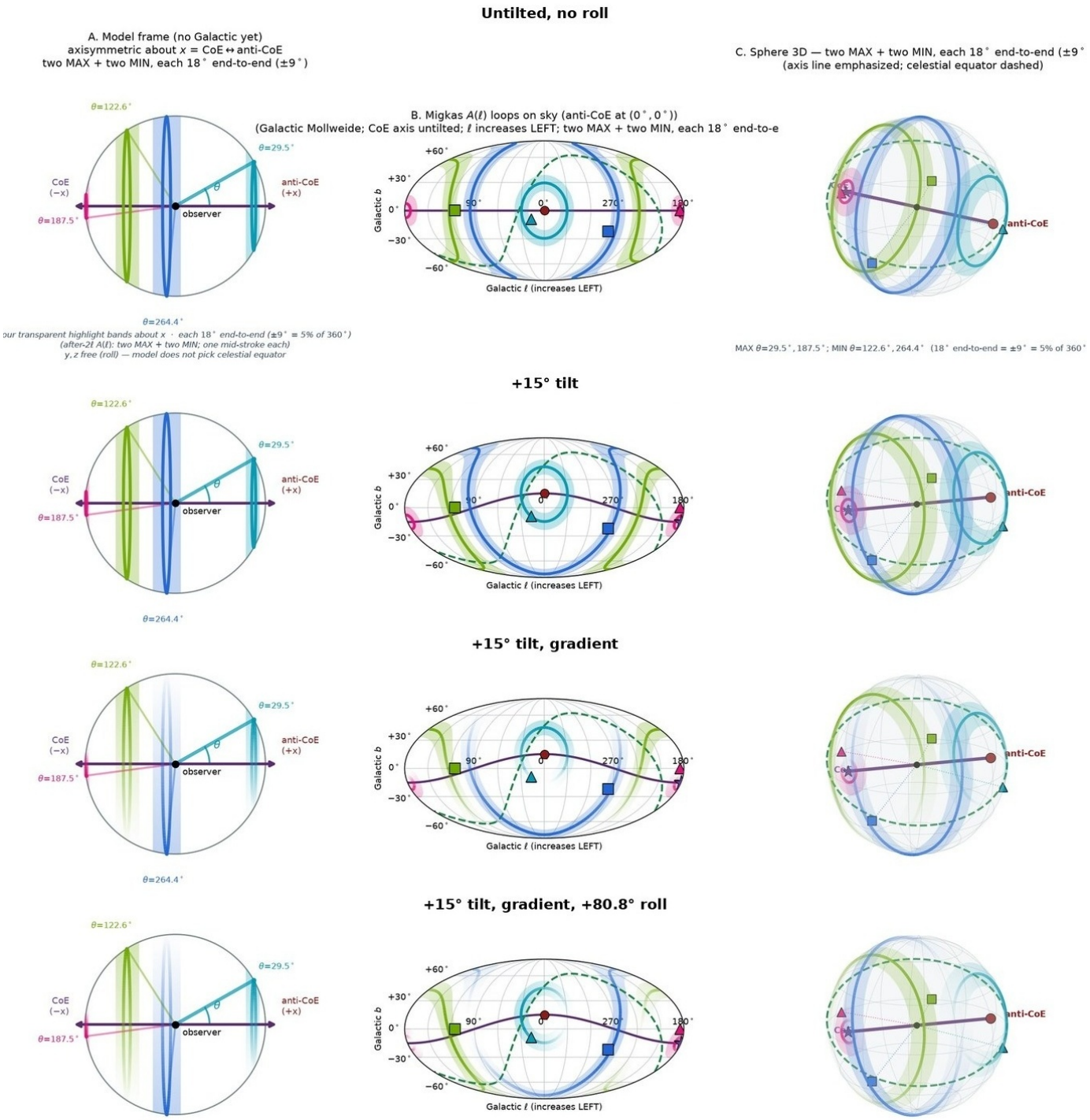


Figure 7: Sky-orientation sequence: untilted configuration, then +15° tilt, radial gradient, and +80.8° roll about the CoE axis. Restored from the original preprint (its Fig. 5).

with $X = M$ or R_v, thinning $f = 1/100$, and $m(\mathbf{x})$ the CoE modulation of Fig. 16. The painted ρ is mean-subtracted inside the region (zero net mass), then Φ is found by FFT Poisson (§5.6).

The kernel ladder: what "kernel at R" means. From `realweb/build.py` (`skyfrom`) and `mc/mc.py`, `mc_samples.py`. *Step 1, acceleration*: $\mathbf{a} = -\nabla\Phi$ on the grid, from Poisson in Fourier space, $\mathbf{a}_k = -i\mathbf{k}\,(4\pi G\rho_k/k^2)$, with a Gaussian of cell width applied to represent finite body size. *Step 2, radial tidal signal on a sphere*: for each HEALPix direction $\hat{n}$ ($N_{\rm side} = 32$, 12,288 directions) and each radius $R \in \{0.5, 0.9, 2, 4, 8, 14\}$ Gpc,

$$s_R(\hat{n}) = \big[\mathbf{a}(R\hat{n}) - \mathbf{a}(\mathbf{0})\big] \cdot \hat{n} - \langle\cdot\rangle_{\rm sky}. \tag{13}$$

This is the radial component of the acceleration at distance R *relative to the observer's own acceleration*: a tidal (differential) field, since a uniform acceleration is

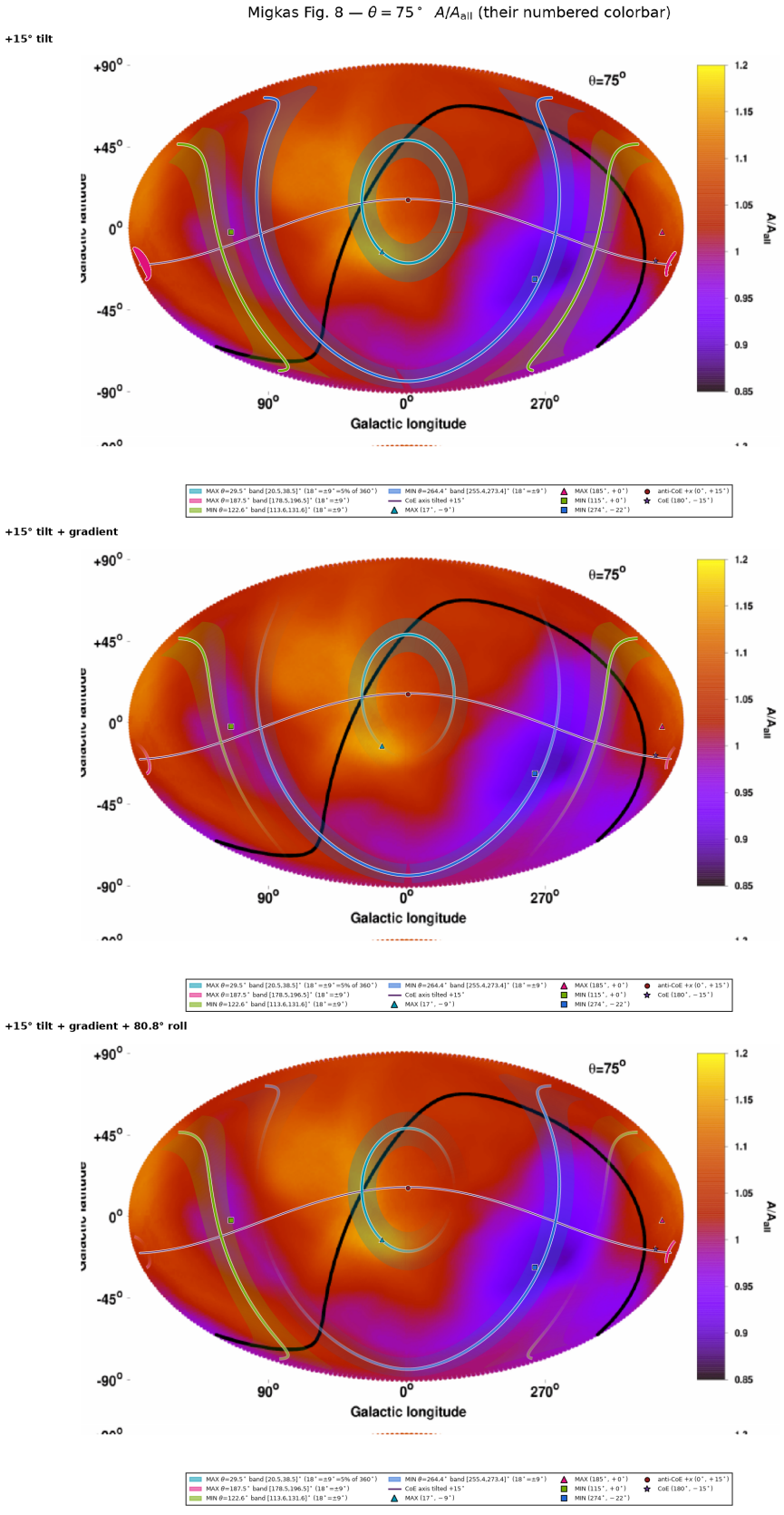


Figure 8: Predicted MAX/MIN bands on Migkas Fig. 8 (two MAX and two MIN bands, each 18° wide) for the +15° tilt alone (top), tilt with gradient (middle), and tilt, gradient and +80.8° roll (bottom). Regenerated from code (the original preprint's Fig. 6).

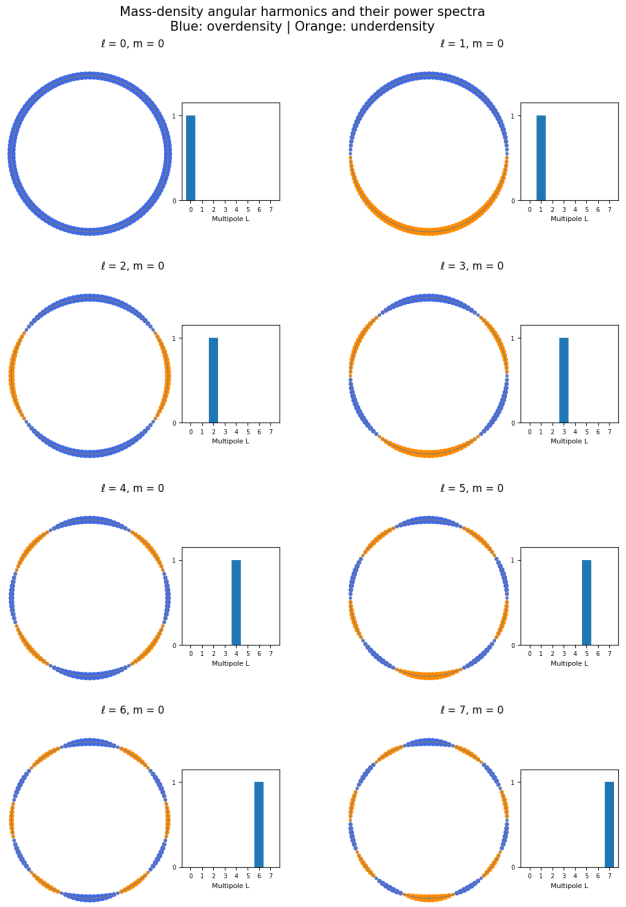


Figure 9: Single axisymmetric harmonics $\ell = 0 \ldots 7$ drawn as density on a ring (blue over-, orange under-density) with the power bar at that ℓ.

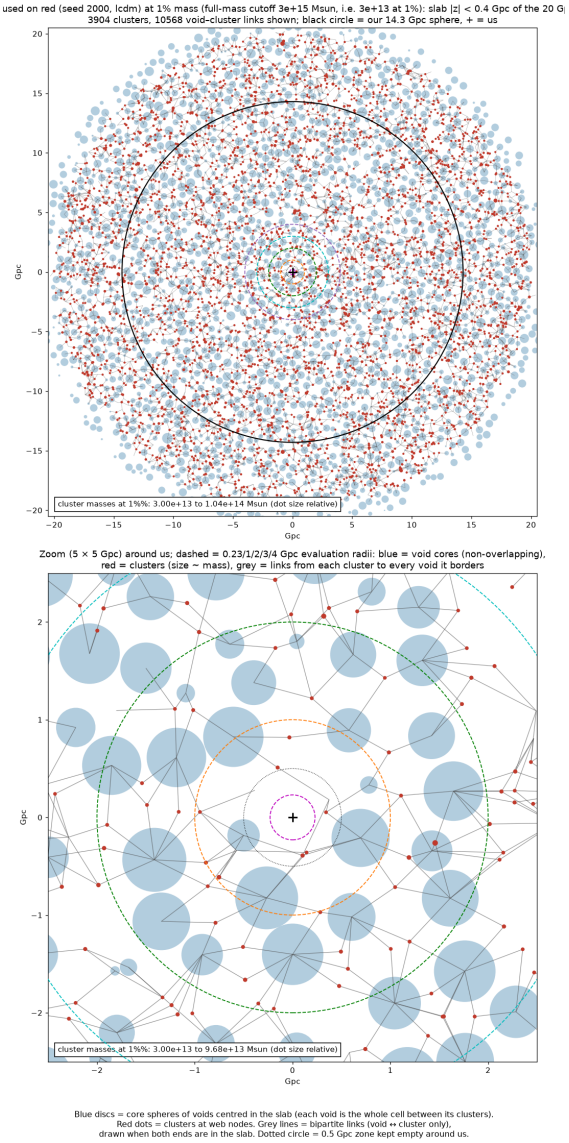


Figure 10: Realistic web: slab $|z| < 0.4\,\mathrm{Gpc}$ of a 20 Gpc box (seed 2000, 1% mass cut, 3904 clusters, 10568 void–cluster links). Circle: 14.3 Gpc sphere; +: observer. Bottom: zoom (±9 Gpc). [unverified: whether this figure used the same `build.py` as above (whose grids use seeds 11, 12)].

unobservable. "Kernel at R Gpc" therefore means that the web's sky imprint is evaluated on the sphere of radius R. Small R weights nearby structure; large R samples the field near the source surface. The map is multiplied by $M_\odot/K_G \cdot R_V^2$ ($K_G = 7.7 \times 10^{51}/0.0039$, $R_V = 14.3$) to put it in the same dimensionless units as the deterministic "red" map. [unverified: the physical origin of the constant K_G]. *Step 3, Monte-Carlo universe*: each universe has a new Gaussian $\delta_{\rm lin}$ and Poisson halos in 8 mass bins (voids omitted in the fast variant, $N = 256$ grids, seeds $500000 + i$). *Step 4, the % levels*: the sky analysed is

Reading Fig. 11. Coloured curves: D_ℓ of $\Theta_{\rm red} + \epsilon s_R$ for the listed ϵ; magenta: best ϵ; red dashed: smooth red alone; black line: ΛCDM ($\chi^2 = 20.1$); points: Planck. The title colours and "my guess" labels come from the plotting script.

$$\Theta_\epsilon(\hat{n}) = \Theta_{\rm red}(\hat{n}) + \epsilon\, s_R(\hat{n}), \tag{14}$$

with $\Theta_{\rm red}$ the smooth axisymmetric CoE background (`sm_red`, $\Theta_{\rm red} = e^{-S(\psi)}/\langle e^{-S}\rangle - 1$, with S a spline in angle ψ from the axis). For each $\epsilon \in \{100, 50, 20, 10, 5, 2, 1, 0.5, 0.2, 0.05\}\%$ the code computes D_ℓ via Eq. (4) and χ^2 via Eq. (6). So "10%" means the web field is scaled to one tenth of its natural (strength-1) amplitude before being added to the background; since D_ℓ is quadratic in the map, its web part scales as ϵ^2. The "best" curve minimises χ^2 over 60 log-spaced ϵ from $10^{-3.5}$ to $10^{0.3}$ (0.03%–200%). *Reading the ladder*: as $\epsilon \to 0$ the curves converge on the smooth-red spectrum (dashed red, $\chi^2 = 471.4$ in the figure legends), whose power sits in odd ℓ because it is axisymmetric. Increasing ϵ fills in the even ℓ and raises the overall level.

Labels on Fig. 11. The panel titles carry "my guess: PASS/FAIL" labels written by `mc_samples.py`. Its rule is: PASS if $\chi^2_{\rm best} \le 100$, |slope of $\ln D_\ell$ vs ℓ over ℓ 4–30| < 0.05, and $\max D_{2..4}/\max D^{\rm Planck}_{2..4} < 2.5$. These are the script's labels, reproduced in the figure; no ver-

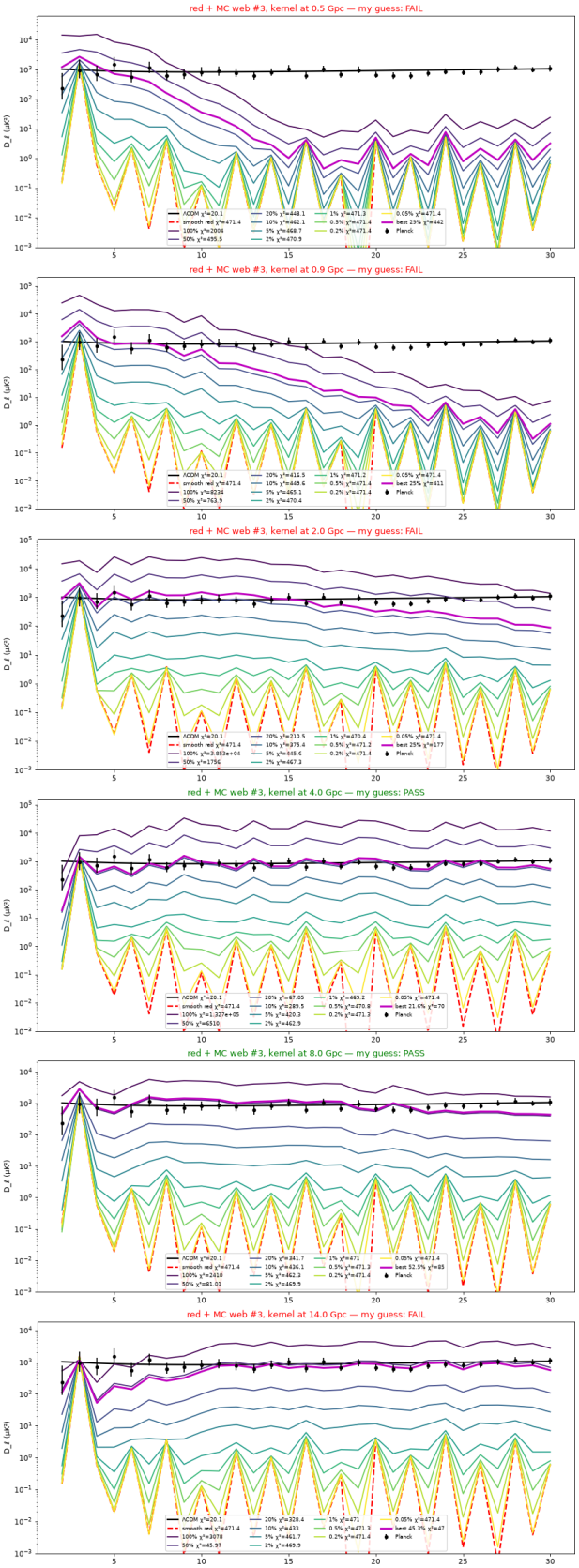


Figure 11: Red + Monte-Carlo web #3, kernel $R = 0.5$, 0.9, 2.0, 4, 8, 14 Gpc (top to bottom); key in text.

dict is drawn here. Recorded values for universe #3 (`samples/guesses.json`), as R: $\epsilon_{\rm best}$, χ^2: 0.5: 29.0%, 441.6; 0.9: 25.0%, 411.5; 2: 25.0%, 177.2; 4: 21.6%, 69.6; 8: 52.5%, 85.0; 14: 45.3%, 47.2. [unverified: at 8 Gpc the JSON records the low-ℓ ratio 2.99, which exceeds the coded 2.5 threshold, yet the label is PASS: the JSON and the script as now written disagree].

5.6 Stochastic field and sky projection

A random field predicts only an ensemble mean C_ℓ, so the residual between one realisation and the mean is expected and is bounded by cosmic variance; only residuals beyond that band are informative.

Why a random field. Instead of placing objects, the density contrast $\delta(\mathbf{x}) = \rho/\bar\rho - 1$ is treated as a zero-mean Gaussian random field. In Fourier space $\langle \delta_{\mathbf{k}} \delta^*_{\mathbf{k}'} \rangle = (2\pi)^3 \delta^3(\mathbf{k}-\mathbf{k}') P_\delta(k)$, where P_δ [Gpc^3] is the variance per mode. The model predicts the *ensemble mean* C_ℓ; one sky scatters about it by Eq. (5).

Poisson equation. Newtonian gravity: $\nabla^2\Phi = 4\pi G\bar\rho\,\delta$. Fourier transforming ($\nabla \to i\mathbf{k}$, so $\nabla^2 \to -k^2$),

$$\Phi_{\mathbf{k}} = -\frac{4\pi G\bar\rho}{k^2}\delta_{\mathbf{k}} \;\Rightarrow\; P_\Phi = \frac{(4\pi G\bar\rho)^2}{k^4} P_\delta. \qquad (15)$$

Code units: $G = 4.30 \times 10^{-12}\,\mathrm{Gpc(km/s)^2}M_\odot^{-1}$ and $\bar\rho = 4.215 \times 10^{19} M_\odot \mathrm{Gpc}^{-3}$, so $4\pi G\bar\rho = 2.278 \times 10^9\,(\mathrm{km/s})^2\mathrm{Gpc}^{-2}$. Example: a mode with $k = 1\,\mathrm{Gpc}^{-1}$ and $\delta_k = 10^{-4}$ gives $|\Phi| = 2.28 \times 10^5\,(\mathrm{km/s})^2$, i.e. $\Phi/c^2 = 2.5 \times 10^{-6}$. Meaning: the k^{-2} makes the potential dominated by large scales.
Validity. Valid for $|\Phi|/c^2 \ll 1$, non-relativistic sources, and scales well inside the horizon ($kR \gg 1$). At $\ell \lesssim 3$, where $k \sim 1/R$, relativistic (GR) corrections are not negligible. The code also evaluates the moving lens with the full CoE radial velocity, which is superluminal ($\dot R \approx 4c$ today per the stage-A notes); the Newtonian treatment of that regime is a model assumption, not a derived result.

Source terms and the moving lens. A photon crossing a potential picks up (i) Sachs–Wolfe $\Theta = \Phi/3c^2$ at the source, (ii) integrated Sachs–Wolfe $\frac{2}{c^2}\int \dot\Phi\, dt$ if Φ evolves, and (iii) the moving-lens term. If the lens moves with transverse velocity $\mathbf{u}$, the potential the photon sees changes during the crossing, giving $\Theta \propto \frac{2}{c^2}\int d\chi\,(\text{time per } d\chi)\,\mathbf{u}\cdot\nabla\Phi$. In `fwd.py` this is the kernel $K_M(\chi) = \frac{2}{c^2}\big[-\frac{\bar A}{a}\frac{\Delta t}{\Delta\chi}\,42\,R_d\big]$ multiplying $\mathbf{u}\cdot\nabla\Phi_0$ [(km/s)^2/Gpc]. Here $\bar A$ is the growth factor D/a, $\Delta t/\Delta\chi$ the time spent per unit path, $42\,R_d$ the CoE radial-velocity factor tabulated from `flowR.py`, and χ the distance from the observer. The stage-A README reports that the moving lens is $\geq 99.99\%$ of the field's power (SW+ISW $\leq 9\times10^{-5}$), so only K_M is kept in the Limber part. [unverified: the closed form of R_d; it is taken from tabulated `tab.npz`].

The $k^2/3$ factor. $\mathbf{u}\cdot\nabla\Phi \to i(\mathbf{k}\cdot\mathbf{u})\Phi_{\mathbf{k}}$. Averaging over random orientations of $\hat k$ relative to $\mathbf{u}$: $\langle(\hat k\cdot\hat u)^2\rangle =$

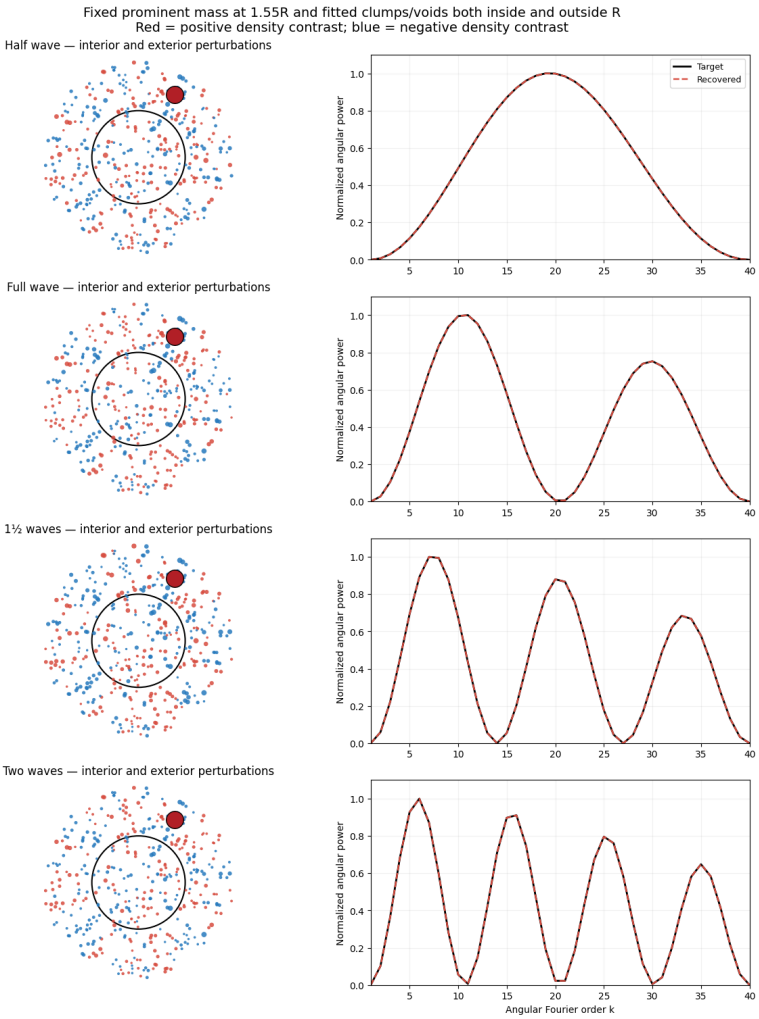


Figure 12: Target profiles matched numerically: a fixed mass at $1.55R$ plus fitted clumps/voids reproduce target angular profiles (half, $1\frac{1}{2}$, $2\frac{1}{2}$, two waves).

$\frac{1}{4\pi}\int \cos^2\theta\, d\Omega = \frac{1}{3}$, so the power is multiplied by $u^2k^2/3$, with u absorbed into K_M.

Radial window. The field is confined to a shell, $\Phi(\chi\hat{n}) = W(\chi)\phi_g$, with

$$W(\chi) = \exp\Big[-\frac{(\chi - R_c)^2}{2w^2}\Big], \tag{16}$$

centre R_c and width w [Gpc]. *Why Gaussian*: two parameters, smooth (no edge artefacts in ℓ), and its derivative $W' = -W(\chi - R_c)/w^2$ is analytic. That derivative is needed because $\mathbf{u}\cdot\nabla(W\phi) = W\,\mathbf{u}\cdot\nabla\phi + \phi\,W'(\hat{n}\cdot\mathbf{u})$. Fitted: $R_c = 3.5715$, $w = 0.3355\,\mathrm{Gpc}$ with $R = 4$. The source surface at 4 Gpc is $(4 - 3.5715)/0.3355 = 1.28w$ above the centre, so the path truncates the Gaussian. That truncation is why Limber needs correcting (below).

Validity. A Gaussian is a parametrisation, not a derived shell profile. It fails to represent sharp-edged or multi-shell structures.

Exact projection (derivation). *Step 1*: the line-of-sight integral is $\Theta(\hat{n}) = \int d\chi\, K(\chi)W(\chi)\,\Phi(\chi\hat{n})$. *Step 2*: Fourier expand, $\Phi(\chi\hat{n}) = \int \frac{d^3k}{(2\pi)^3}\Phi_{\mathbf{k}}e^{i\mathbf{k}\cdot\hat{n}\chi}$, and use the plane-wave expansion $e^{i\mathbf{k}\cdot\hat{n}\chi} = 4\pi\sum_{\ell m} i^\ell j_\ell(k\chi)Y_{\ell m}(\hat{n})Y^*_{\ell m}(\hat{k})$. *Step 3*: read off $a_{\ell m} = \frac{i^\ell}{2\pi^2}\int d^3k\,\Phi_{\mathbf{k}}Y^*_{\ell m}(\hat{k})\Delta_\ell(k)$ with $\Delta_\ell(k) = \int d\chi\, KW j_\ell(k\chi)$. *Step 4*: take $\langle|a_{\ell m}|^2\rangle$ with the power-spectrum definition. The δ^3 collapses one k integral and $\int d\Omega_k|Y_{\ell m}|^2 = 1$, giving

$$C_\ell = \frac{2}{\pi}\int k^2dk\, P_\Phi(k)\,|\Delta_\ell(k)|^2. \tag{17}$$

j_ℓ is the spherical Bessel function: the radial profile of a plane wave's ℓ-th multipole. For the moving lens, $P_\Phi \to \frac{k^2}{3}P_\Phi$, and `fwd.py` evaluates the W' term exactly with $\cos\theta$ Gaunt couplings to $\ell \pm 1$. This is used for $\ell \le 50$.

Limber approximation (derivation). For large ℓ, $j_\ell(k\chi)$ is sharply peaked at $k\chi \approx \nu \equiv \ell + \frac{1}{2}$. Using the closure relation $\int k^2dk\, j_\ell(k\chi)j_\ell(k\chi') = \frac{\pi}{2\chi^2}\delta(\chi - \chi')$, and assuming P_Φ varies slowly over the width of j_ℓ so it can be evaluated at $k = \nu/\chi$, Eq. (17) collapses to a single

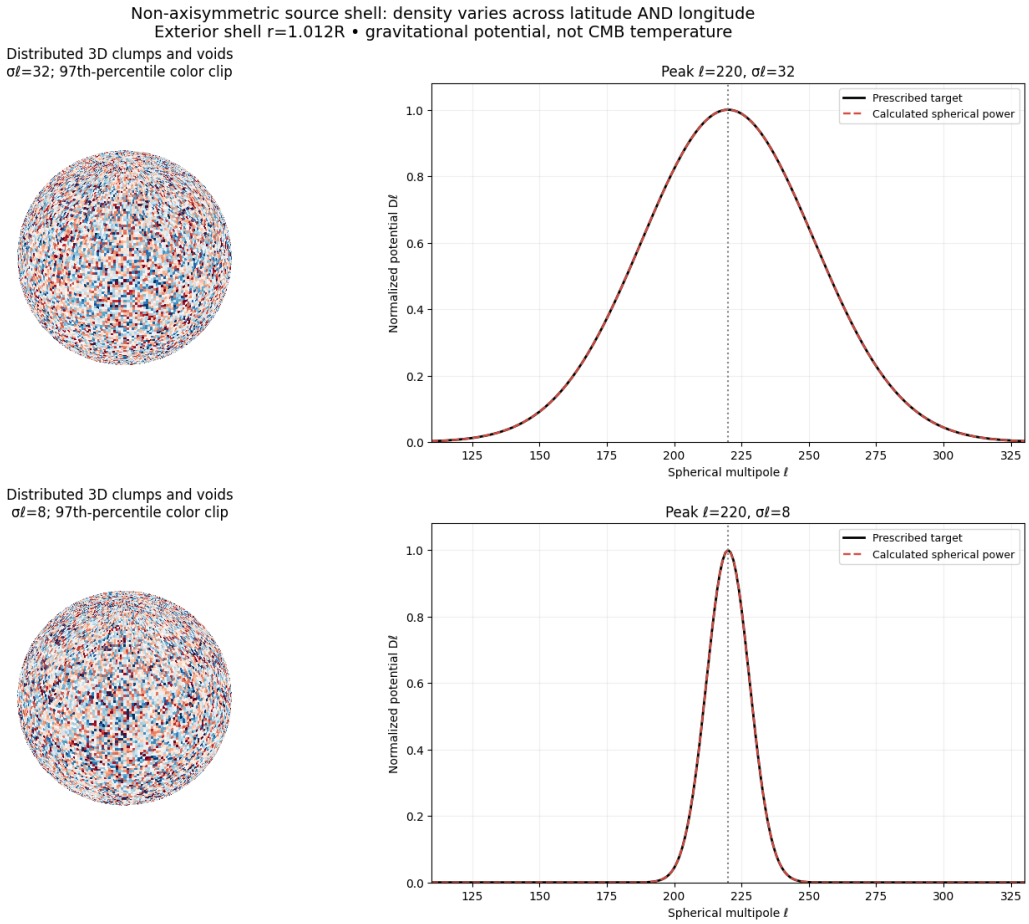


Figure 13: Target peak matched numerically: a non-axisymmetric source shell at $r = 1.012R$ (gravitational potential, not temperature) built from 3D clumps/voids reproduces a prescribed peak at $\ell = 220$ with widths $\sigma_\ell = 32$ and 8.

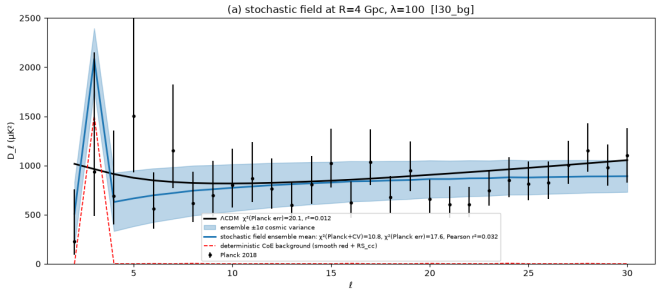


Figure 14: First stochastic-field fit (stage A: Gaussian random field riding the CoE flow, $R = 4\,\mathrm{Gpc}$, ℓ 2–30, CoE background included, $\lambda = 100$). Blue: ensemble mean with $\pm1\sigma$ cosmic-variance band ($\chi^2_{\mathrm{Planck+CV}} = 10.8$, $\chi^2_{\mathrm{Planck}} = 17.6$, Pearson $r^2 = 0.032$); red dashed: deterministic CoE background; black: ΛCDM ($\chi^2 = 20.1$, $r^2 = 0.012$); points: Planck 2018.

integral:

$$D_\ell \simeq \frac{\ell(\ell+1)T_0^2}{2\pi}\int d\chi \times \frac{[K_M W]^2}{\chi^2}\frac{k^2}{3}P_\Phi(k)\Big|_{k=\nu/\chi}. \tag{18}$$

Meaning: each multipole receives power only from the wavenumber that projects to angle $1/\ell$ at distance χ. Only transverse modes contribute; radial modes cancel along the line of sight.

Validity. It requires the kernel $K_M W$ to vary slowly compared with j_ℓ's oscillation $\sim \chi/\nu$, i.e. $\ell \gg R_c/w = 3.5715/0.3355 = 10.6$, and no sharp edges. Checks in `acoustic_limber_check*.json`: interior or broad shells agree within 3% for $\ell \geq 150$, and a 0.5 Gpc shell within 0.3% even at $\ell = 20$. Shells cut off at the path end (as here) are overestimated by about 1.5–5× below $\ell = 150$ and by up to ±20% near $\ell = 300$.

Exact/Limber correction. Hence $D_\ell^{\mathrm{used}} = c_\ell D_\ell^{\mathrm{Limber}}$ for $\ell > 50$, with $c_\ell = C_\ell^{\mathrm{exact}}/C_\ell^{\mathrm{Limber}}$ computed by Eq. (17) at 69 nodes. The nodes are $\ell = 50, 55, 60, 70, 80, 90, 100, 115, 130, 150, 175$, then 200 to 2480 in steps of 40. c_ℓ is interpolated in $\ln\ell$ and recomputed after each fit pass, because it depends on R_c and w.

Fitted source spectrum.

$$P_\delta = 10^{\lg A} k^n e^{-(k/k_D)^2}\left[1 + A_w\cos(kr_s + \phi) + Be^{-k/k_B}\cos(kr_2 + \phi_2)\right], \tag{19}$$

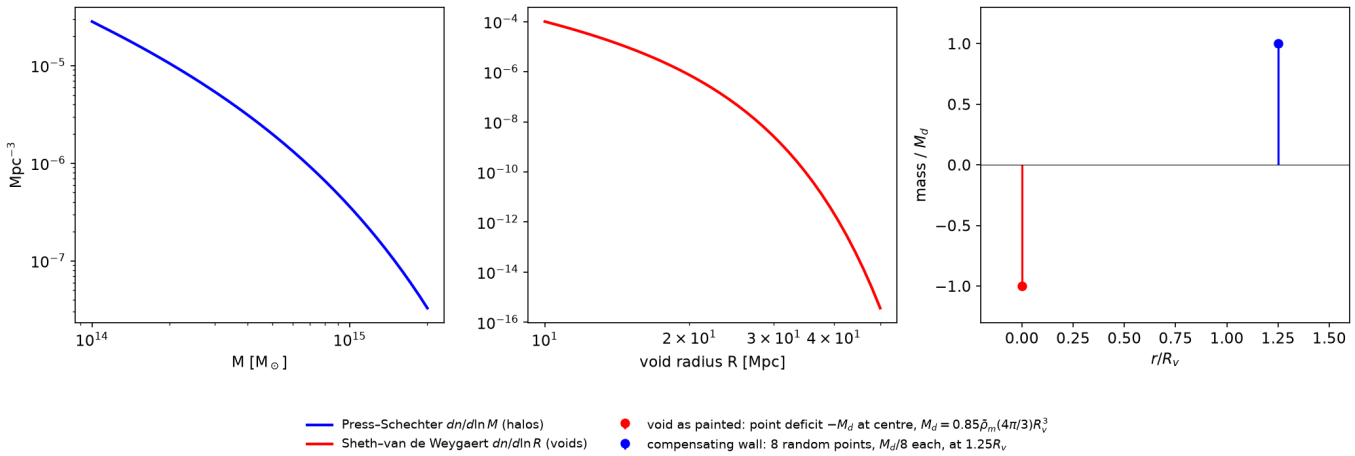


Figure 15: Web ingredients recomputed from `cosmo.py`: halo mass function (left), void size function (middle), compensated void mass placement (right).

with k in Gpc^{-1} and P_δ in Gpc^3. Term by term: k^n is the broadband tilt; $e^{-(k/k_D)^2}$ is a small-scale damping cutoff; $A_w \cos(kr_s + \phi)$ is an oscillation in k with period $2\pi/r_s$, i.e. a preferred comoving separation r_s (acoustic-like); the B term is a second, decaying oscillation. Fitted (r_2 free, `acoustic_half_fixed.json`): $\lg A = -8.282$, $n = -0.183$, $A_w = 0.311$, $r_s = 79.5\,\mathrm{Mpc}$, $\phi = 1.367$, $k_D = 335\,\mathrm{Gpc}^{-1}$, $B = 0.815$, $k_B = 138\,\mathrm{Gpc}^{-1}$, $r_2 = 42.1\,\mathrm{Mpc}$, $\phi_2 = 3.640$. Example: at $k = 100\,\mathrm{Gpc}^{-1}$, $10^{-8.282} \times 100^{-0.183} \times e^{-0.089} = 2.2 \times 10^{-9}\,\mathrm{Gpc}^3$ before the bracket.

Mapping $\Delta\ell \leftrightarrow r$ via $\ell \approx k\chi$. From Limber, a wavenumber k at distance χ appears at $\ell \approx k\chi$. An oscillation $\cos(kr_s)$ has period $\Delta k = 2\pi/r_s$, so on the sky its peaks are spaced by

$$\Delta\ell \approx \chi_{\mathrm{eff}} \Delta k = 2\pi\chi_{\mathrm{eff}}/r_s. \tag{20}$$

Example: acoustic fit, $R_c = 3.944\,\mathrm{Gpc}$, $r_s = 0.0819\,\mathrm{Gpc}$, gives $\Delta\ell = 302.6$; the model peaks 227, 530, 831, 1131, 1431 are spaced 300–303. For the final fit, $R_c = 3.5715$ and $r_s = 0.0795$ give 282, whereas the model peaks are spaced ≈ 300. That implies $\chi_{\mathrm{eff}} = 300 \times 0.0795/2\pi \approx 3.8\,\mathrm{Gpc}$, larger than R_c because K_M rises toward the source and the truncated window is skewed. Conversely, an angular feature at ℓ has physical size $\chi\theta$ with $\theta = \pi/\ell$. For $\ell = 220$: 57.1, 114.2, 199.9 Mpc at $\chi = 4$, 8, 14 Gpc. *Validity.* It holds where Limber holds and for thin kernels. It fails at low ℓ and for broad kernels, where many χ contribute and χ_{eff} is ambiguous.

5.7 The fixed low-ℓ background D_ℓ^{fix}

The deterministic, axisymmetric part of the CoE sky is modelled out once as D_ℓ^{fix}, so that the later fits work on the residual above this background.

Definition (from `fit.py`). The CoE sky has a deterministic, axisymmetric part: the smooth red map plus an `rs_cc` map. Its spectrum is computed once,

$$D_\ell^{\mathrm{fix}} = \frac{\ell(\ell+1)}{2\pi} T_0^2 \, \hat{C}_\ell \big[\Theta_{\mathrm{red}} + \Theta_{\mathrm{rs_cc}}\big], \quad 2 \le \ell \le 30, \tag{21}$$

with the dipole removed, and stored in `fit_l30_bg.json`. It is added to the field: $D_\ell^{\mathrm{model}} = c_\ell D_\ell^{\mathrm{field}} + D_\ell^{\mathrm{fix}}$. *Why add powers*: the field has zero mean and is independent of the background, so $\langle|a^{\mathrm{bg}} + a^{\mathrm{f}}|^2\rangle = |a^{\mathrm{bg}}|^2 + \langle|a^{\mathrm{f}}|^2\rangle$ (the cross term averages to zero). *Values*: $\ell = 3$: $1506.4\,\mu\mathrm{K}^2$; $\ell = 2$: 0.1; all others ≤ 5.1, with odd ℓ (5, 7, …) carrying 0.3–5.1 and even $\ell \ge 4$ essentially 0. The odd/even pattern is the axisymmetric signature: an odd-parity profile about the axis projects onto odd Legendre polynomials. It is held fixed (no free parameters) and has no cosmic-variance band, since it is deterministic. Above $\ell = 30$ it is set to zero; its power is already negligible above $\ell = 3$. [unverified: the physical definition of `rs_cc`].

5.8 Acoustic fit and residual decomposition

After the fixed background, an acoustic stochastic term is fitted. Its residual against Planck is still large (the χ^2 and r^2 below are worse than ΛCDM), and the oscillatory structure left in that residual is what the damped model of the next section is built to describe.

Acoustic fit. Eq. (19) without the B term, eight free parameters with R_c and w free ("free_W"). Result: $r_s = 81.9\,\mathrm{Mpc}$, $R_c = 3.944\,\mathrm{Gpc}$; $\chi^2_{2-2500} = 11{,}772$, Pearson $r^2 = 0.871$ (ΛCDM: 2,558, 0.983). Peaks: CoE 227, 530, 831, 1131, 1431 vs Planck 224, 535, 821, 1122, 1412.

Residuals. Normalised residual: $\epsilon_\ell = (D_\ell^{\mathrm{P}} - D_\ell^{\mathrm{CoE}})/\sigma_\ell$, dimensionless (in σ). Two linear models with $s_j(\ell) = \sin(2\pi j\ell/\Delta\ell + \phi_j)$:

$$\text{undamped: } \epsilon_\ell = \textstyle\sum_j a_j s_j(\ell), \tag{22}$$

$$\text{damped: } \epsilon_\ell = e^{-\ell/L} \textstyle\sum_j a_j s_j(\ell). \tag{23}$$

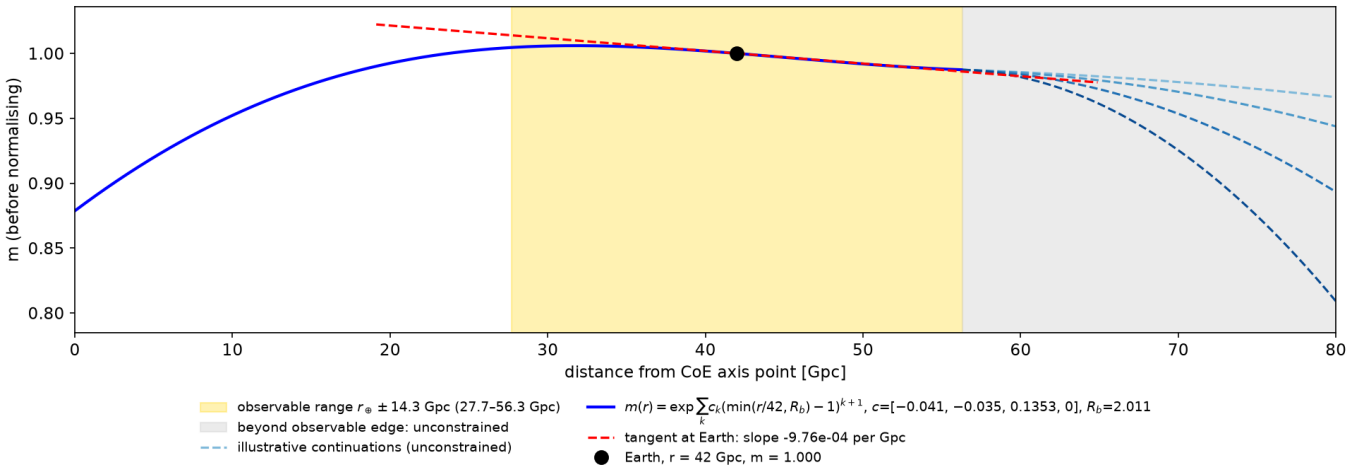


Figure 16: CoE modulation $m(r) = \exp\sum_k c_k(\min(r/42, R_b) - 1)^{k+1}$ of the web number density. r is measured from the axis point $42\hat{a}$ Gpc, $\hat{a} \propto (1, 0, \tan 15°)$; $c = [-0.041, -0.035, 0.1353, 0]$, $R_b = 1.5(1 + 1/2.937)$. Normalised to mean 1 in each grid shell. Black dot: Earth at $r = 42$ Gpc ($m = 1$); red dashed: tangent there, slope -9.8×10^{-4} Gpc^{-1} (cf. CatWISE-locked $\rho'/\rho \approx -0.00106$); gold band: how far we see in either direction, $r = 42 \pm 14.3$ Gpc ($d = 14.3$ Gpc, the $D \pm d$ reach of §5.1). Grey ($r > 56.3$ Gpc): assumed uniform at the band-edge value $m = 0.987$; dotted: formula continued, not used.

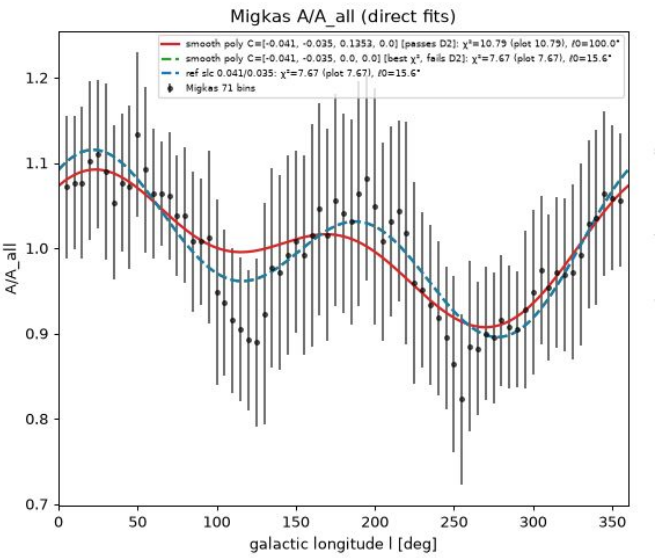


Figure 17: Migkas $A/A_{\rm all}$ (71 bins, black) with CoE direct fits. Red: smooth polynomial in $\cos\psi$, $\chi^2 = 10.79$, $l_0 = 100°$. Green dashed: best variant, $\chi^2 = 7.67$, $l_0 = 15.6°$. Blue dashed: reference sinusoid (amplitudes 0.041/0.035), $\chi^2 = 7.67$. The two humps come from $\cos\psi(l)$ at fixed galactic latitude. Red uses the c_k of $m(r)$, Fig. 16.

Why harmonics: if the model's peak spacing or shape is slightly off, the residual is periodic in ℓ with the spacing error. $j = 2$ is the half period, which is the signature of an odd/even peak alternation. *Why damping*: σ_ℓ shrinks and the signal falls with ℓ, so the misfit in σ units decays. A single envelope $e^{-\ell/L}$ adds one parameter. *Fit*: for fixed $(\Delta\ell, L)$ the model is linear in $a_j \cos\phi_j$, $a_j \sin\phi_j$ (least squares), and $\Delta\ell$, L are scanned. Results: undamped plateaus at $r^2 = 0.551$; damped, 1–9 harmonics give $r^2 = 0.797$, 0.813, 0.831, 0.834, 0.835, 0.836, 0.836, 0.837, 0.837, with $\Delta\ell \approx 593$ and $L \approx 990$. Starting amplitudes (σ): 34.6, 5.6, 5.7, 2.2, 1.3, 1.0, 1.2, 0.4, 0.5. [unverified: these values were taken from the earlier figure set; the script was not re-run]. The decomposition is purely descriptive. It fails where the residual structure is not stationary, e.g. the late troughs at $\ell \sim 1400$, 2000, which the shared envelope has already suppressed.

Mapping to source parameters. Using Eq. (20) with $\chi \approx 3.94$ Gpc: a residual period $\Delta\ell = 593$ corresponds to $r_2 = 2\pi \times 3.94/593 = 0.0417$ Gpc $= 41.7$ Mpc $\approx r_s/2$. An envelope $e^{-\ell/L}$ with $\ell = k\chi$ becomes $e^{-k\chi/L}$, so $k_B = L/\chi = 990/3.94 = 251$ Gpc$^{-1} = 0.25$ Mpc^{-1}. These were the starting values for the damped term.

Table 2: Fit statistics (Planck errors). Pearson r^2 over ℓ 2–2500.

Model	χ^2_{2-2500}	r^2	χ^2_{2-30}
CoE r_2 free	3,787	0.976	137
CoE half ($r_2 = r_s/2$)	4,131	0.966	49.1
CoE acoustic (no B)	11,772	0.871	–
ΛCDM	2,558	0.983	20.1

peak	Planck	CoE r_2 free	CoE half
1	228 (5726)	218 (5927)	207 (5223)
2	544 (2580)	531 (2449)	537 (2383)
3	823 (2525)	813 (2466)	818 (2433)
4	1121 (1239)	1130 (1293)	1132 (1260)
5	1417 (814)	1420 (822)	1426 (861)

Lower: peak ℓ (D_ℓ in μK^2).

5.9 Damped half-scale model

The second damped mode is a hypothesis taken from the acoustic-fit residual. Here it is added to the model and refitted, and the question is how much of that residual structure it removes.

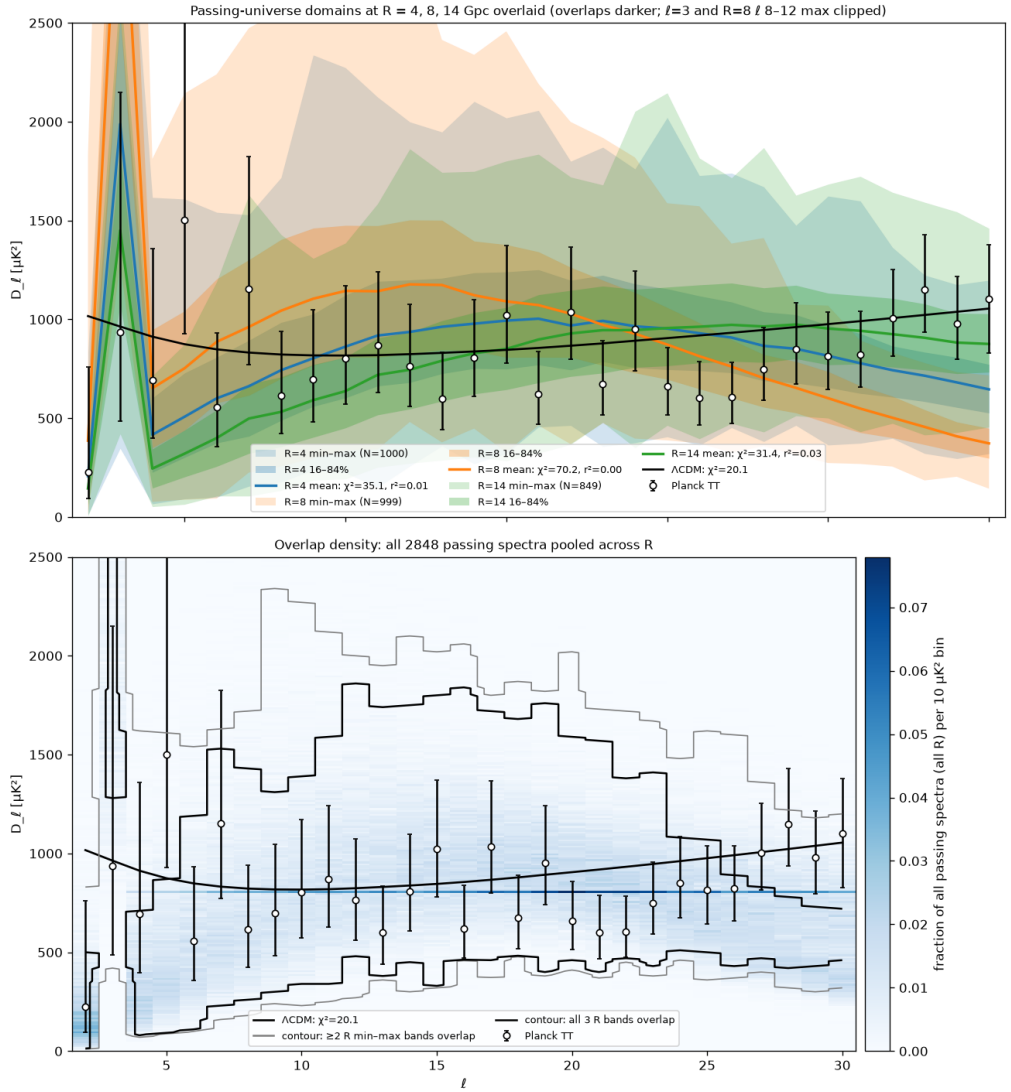


Figure 18: Passing-universe domains at $R = 4$, 8, 14 Gpc overlaid (top; bands are spreads over realisations; $\ell = 3$ and the $R = 8$ ℓ 8–12 maxima clipped) and the density of sky-plane domains (bottom) against Planck.

Model and fit. Eq. (19) with all 12 parameters $(\lg A, n, A_w, r_s, \phi, k_D, R_c, w, B, k_B, r_2, \phi_2)$ floated in one χ^2 minimisation over $\ell = 2$–2500 (Eq. (6)). The "half" variant ties $r_2 = r_s/2$ (11 parameters). Result (r_2 free): $r_2 = 42.1$ Mpc vs $r_s/2 = 39.8$; $k_B = 0.138\,\mathrm{Mpc}^{-1}$, against the starting 0.25. The resulting model is shown in Fig. 24.

The $\ell = 50$ step. A ratio c_ℓ computed for an earlier, narrower shell had been left in place when the fit widened the shell, which produced a step at $\ell = 50$ (reported as 43%). Recomputing c_ℓ on the 69 nodes at each pass reduces the 50→51 change to ≈ 1.1%. [unverified: the 43% and 1.1% figures come from the run log; not re-measured here].

Remaining structure. The odd/even alternation is absent from the new residual. What remains: a slow wave with $\Delta\ell \approx 1{,}384$ (± 8–11σ), and $\ell = 2$–3 raised to $\approx 3{,}300\,\mu\mathrm{K}^2$, which dominates $\chi^2_{2-30} = 137$; part of the $\ell = 3$ level is $D_3^{\mathrm{fix}} = 1506$. [unverified: the $\Delta\ell \approx 1384$ amplitude and the $3300\,\mu\mathrm{K}^2$ level were read from the earlier figures].

5.10 Residual harmonics against independent sky anomalies

The A and B residuals left by the CoE model are peeled layer by layer, $F_1 + F_2 + F_3$ for $n = 1$, 2, 3, each fitted to what the previous layer left. The leftover map is then compared with sky anomalies found independently. Agreement there is qualitative and is reported as such.

5.10.1 Residual decomposition of A and B

Start from the A and B reconstructions (313 clusters, Migkas error bars) minus the CoE model (A: AF_smooth; B: sheet BL). Harmonics $n = 1$, 2, 3 in ℓ are fitted jointly to both residuals, weighted by the Migkas errors, each on

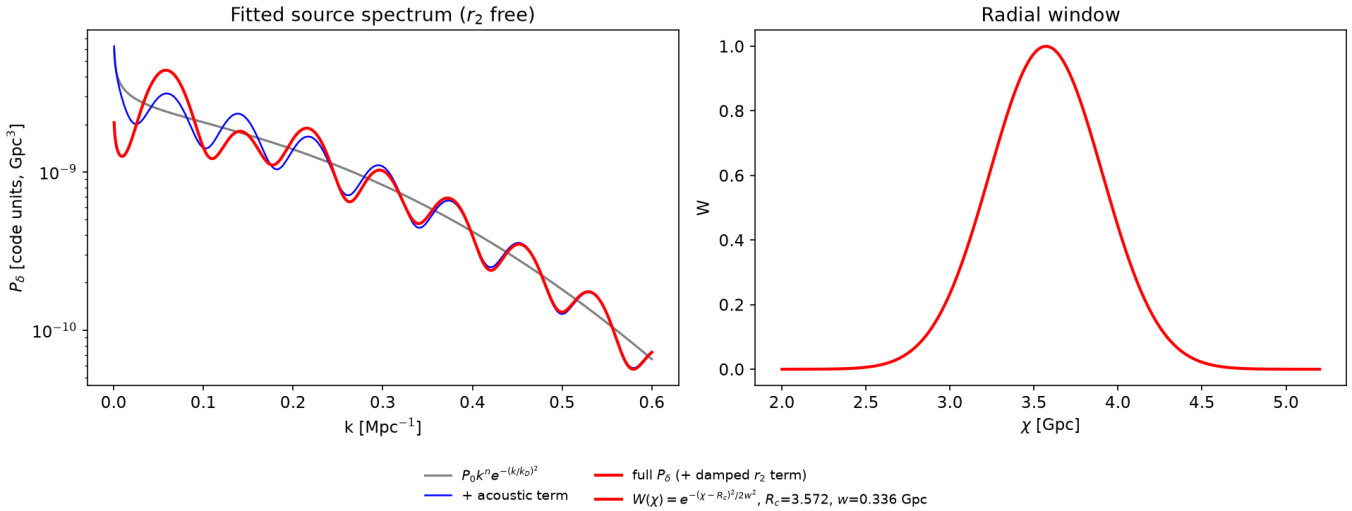


Figure 19: Fitted P_δ split into its terms (left) and the radial window $W(\chi)$ (right).

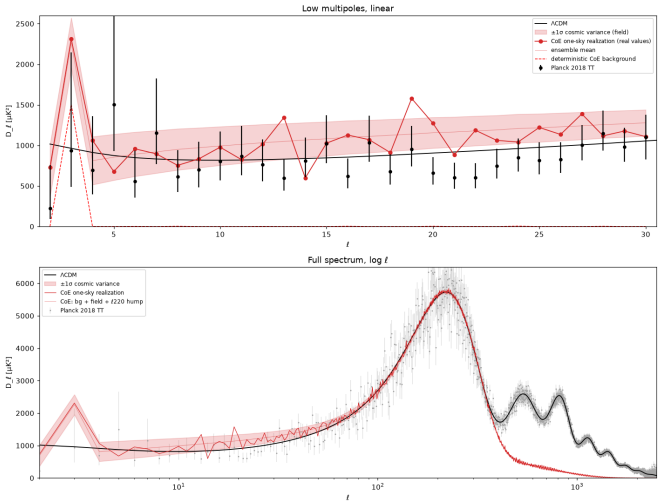


Figure 20: Low multipoles (linear axes): ΛCDM (black), Planck (grey), CoE one-sky realisation (red) in its $\pm 1\sigma$ cosmic-variance band (Eq. (5)) with faint ensemble mean. Bottom: full spectrum.

the residual left by the previous one. Each harmonic has a shared amplitude and phase and a separate offset for A and B. $n = 1$ (amplitude 0.0382, peak at $\ell = 102.8°$) is a dipole in ℓ: it tilts the model and corrects the depth of the A trough near $\ell \approx 100°$. $n = 2$ (0.0140, peaks at 54.5° and 234.5°) is a quadrupole that reshapes the two humps. $n = 3$ (0.0064, peaks at 82.7°, 202.7°, 322.7°) is a small fine correction. For A, χ^2 falls from 25.54 to 15.30, 12.06 and 11.46. For B the values are 13.68, 8.67, 10.04 and 10.24 (Fig. 25). For B, χ^2 is lowest after $n = 1$ and rises slightly with the shared $n = 2$ and 3 terms. The residual used below is 100 (data − model)/data after the three harmonics, averaged over ±25° in ℓ. Code: `dumpM.py`, `recon_n3.py`, `cum2.py`.

5.10.2 Sky anomalies

The A and B residuals after removing the $n = 1$–3 harmonics are drawn on the sky. The model map is lifted 15° (anti-CoE up) and rotated 45° counterclockwise about the CoE axis; observed anomalies stay at their published (ℓ, b). Regions already used in the Migkas fit are excluded. Each spot is tied to A or B by a qualitative physical assignment and scored by sign only: a hot spot needs residual > 0, a cold spot < 0. Result: A 4/5, B 7/7, total 11/12; the CMB dipole cold pole sits on the zero line (+0.02%) and is marked "?". In Fig. 25 the points are the reconstruction from 313 clusters with Migkas Fig. 6 error bars; the CoE model is AF_smooth for A and the sheet BL for B; the shared harmonics come from a joint error-weighted fit to A and B with shared amplitude and phase and separate offsets (`cum2.py`/`recon_n3.py`), giving $C_1 = 0.0382$ (102.8°), $C_2 = 0.0140$ (109.0°), $C_3 = 0.0064$ (248.2°). Residuals after each step are overlaid in the bottom panels with their RMS; shading marks ℓ 15–80° and 260–335°. Bootstrap 1σ per spot is 3–6 percentage points (2000 resamples of the 313 clusters).

Qualitative assignment of spots to A or B. A = normalisation of the L–T relation (distance / expansion rate / velocity). A spot is tied to A when it measures motion or distance along the line of sight.

- CMB dipole (hot and cold poles): our velocity relative to the CMB rest frame, a kinematic quantity.
- Bulk flow (toward and away): peculiar velocities of galaxies and clusters, i.e. recession velocity.
- CMB Cold Spot: associated with the Eridanus supervoid, an underdensity that alters expansion and photon path along that sightline (distance/expansion effect).

B = slope of the L–T relation (sample composition). A spot is tied to B when it reflects source counts, sample makeup or foreground contamination.

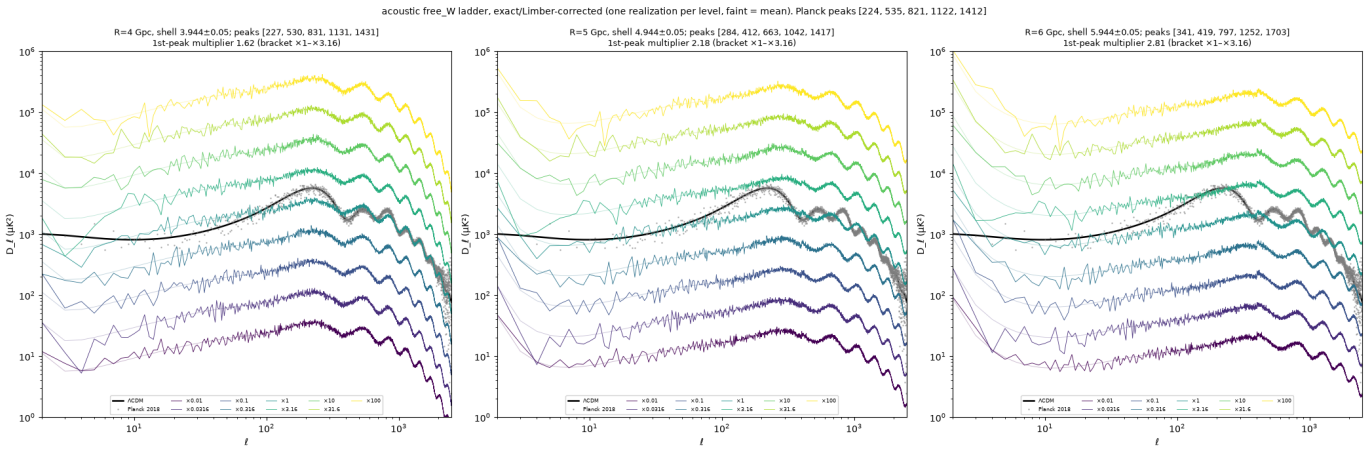


Figure 21: The same source parameters with $R = 4$, 5, 6 Gpc (shell at 3.944, 4.944, 5.944 Gpc): peaks [227, 530, 831, 1131, 1431], [284, 412, 663, 1042, 1417], [341, 419, 797, 1252, 1703]. Moving the shell without refitting r_s changes $\chi_{\rm eff}$ in Eq. (20).

- NVSS and CatWISE dipoles: number-count dipoles of radio sources and quasars, set by how many sources pass a flux cut in each direction (selection/sample effect).

- Hemispherical power asymmetry: a difference in fluctuation amplitude between sky halves, a property of the population rather than a velocity.

- Loop I / North Polar Spur: Galactic foreground changing $N_{\rm H}$ absorption and soft X-ray background, shifting luminosities differently with temperature (slope change).

Assignments were made from the physics before reading any residual sign. The scores for each spot are listed in Fig. 27.

6 Discussion

What is fitted and what is predicted. Fitted: D, the density-law shape and l_0 (cluster data); the 12 source and shell parameters (CMB). Given those, the two-hump shape of $A/A_{\rm all}$, the CatWISE-locked L and the full D_ℓ through Eqs. (17)–(18) follow. The band morphology on the Migkas maps was fixed from the geometry, but its amplitude depends on the normalisation of the construction.

Tensions. The cluster anisotropy is the model's strongest point: a single off-centre axis and a density gradient set by an independent dipole bring χ^2 from 45.14 to about 10–12. The refitted D ranges from 34.96 to 59.8 Gpc depending on the law, so the distance is not well determined. The CMB is the main difficulty. The acoustic fit is far worse than ΛCDM (χ^2 = 11,772 vs 2,558), and even the final model (3,787) does not reach it; the second damped mode the data require, at $r_2 \approx 42$ Mpc

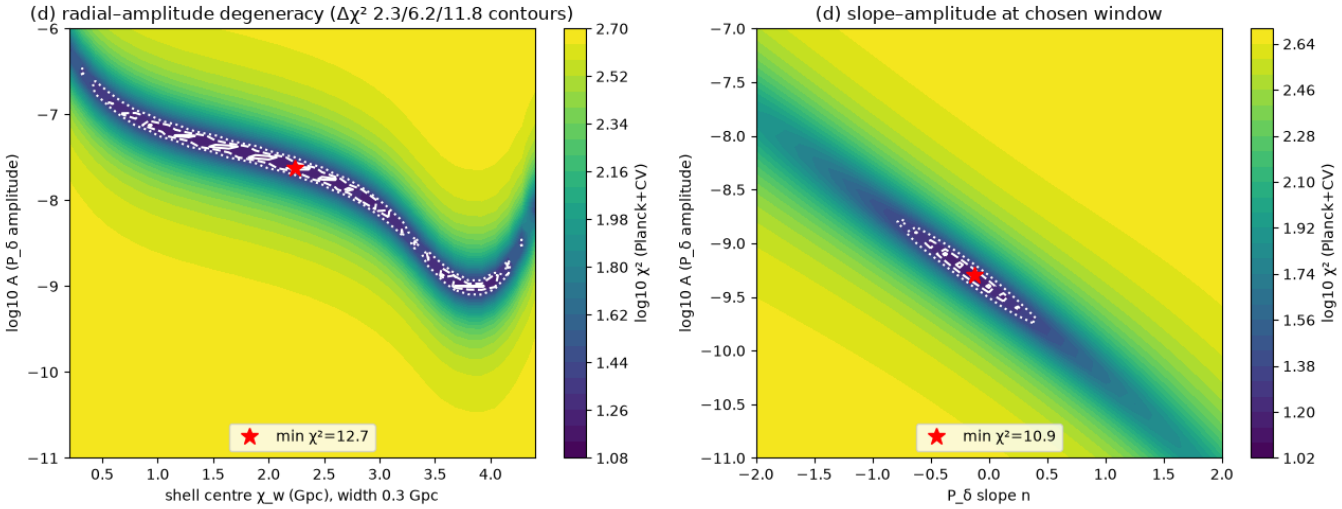


Figure 22: Parameter degeneracies. Left: shell position vs amplitude ($\Delta\chi^2 = 2.3, 6.2, 11.8$, i.e. 1, 2, 3$\sigma$ for two parameters). A farther shell needs a different amplitude because K_M varies with χ. Right: tilt vs amplitude at the chosen window.

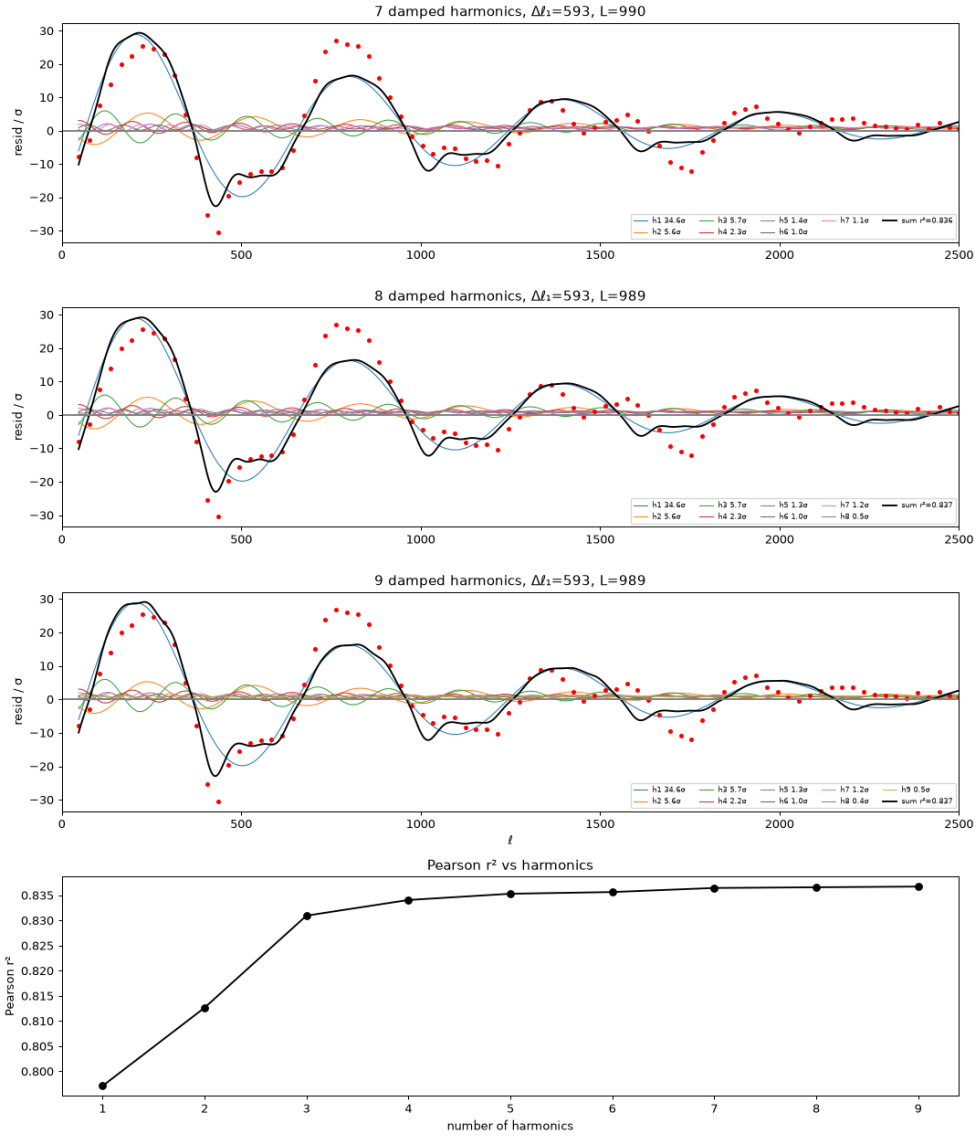


Figure 23: Damped-harmonic fits to ϵ_ℓ with 7, 8, 9 harmonics ($\Delta\ell = 593$, $L = 990$) and Pearson r^2 versus number of harmonics (bottom).

$\approx r_s/2$ decaying at $k_B \approx 0.14\,\mathrm{Mpc}^{-1}$, has no physical origin in the model yet. The low multipoles are too high ($\chi^2_{2-30} = 137$). Two sky orientations (the $+80.8°$ roll and the $45°$ rotation) are used and have not been reconciled.

Limits. The thermal history is homogeneous FRW by assumption; the moving lens is treated in Newtonian terms in a superluminal regime; Limber is inaccurate for the truncated shell below $\ell = 150$; and several numbers are flagged as not re-derived. The 11/12 sky-anomaly agreement is qualitative, with no stated false-positive rate. The density law is constrained only across $D \pm d$; near the CoE and beyond 56.3 Gpc it is unconstrained. Whether an off-centre observer would infer an apparent acceleration from supernova distances has not been computed here.

Next steps. (1) Exact projection throughout, and control of low ℓ in the r_2-free fit. (2) An analytic thin-shell projection to see which parameters the geometry fixes. (3) $D(t)$ from the Newtonian integrator. (4) Reconcile the PASS label at 8 Gpc with the coded rule. (5) Luminosity distances along off-centre lines of sight, to address the acceleration question directly.

7 Conclusions

We have described a toy universe of ordinary matter expanding from a centre under classical laws, seen by an observer off that centre, and compared it with the cluster sky, the CatWISE dipole and the CMB.

- The off-centre geometry reproduces the two-hump, two-trough longitude dependence of the Migkas A/A_{all}: with the density gradient locked to the CatWISE dipole, χ^2 falls from 45.14 to about 10–12, and to 7.67 for the best direct fit.

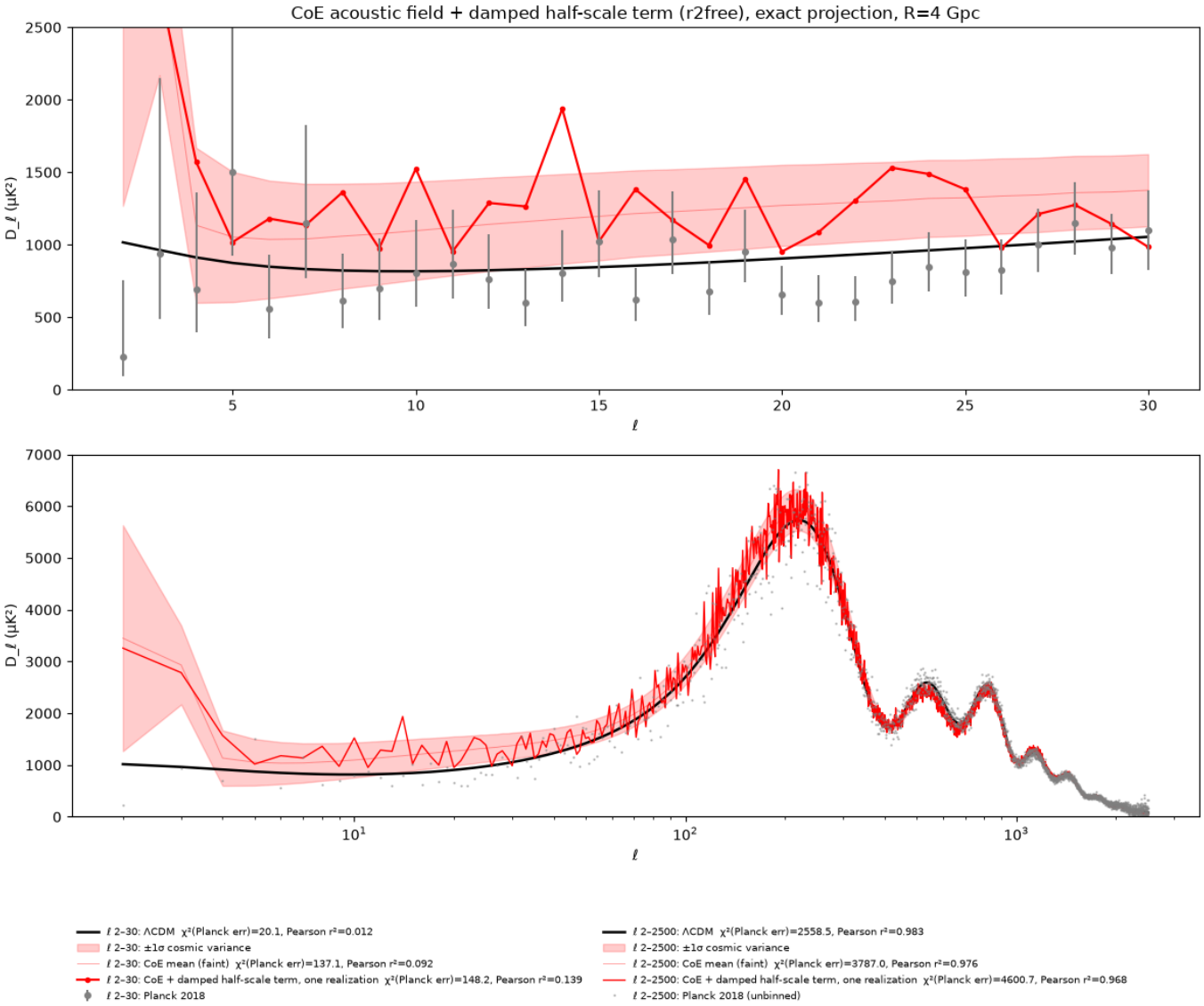


Figure 24: Final model: acoustic field + damped half-scale term (r_2 free), exact/Limber-corrected projection, $R \approx 4\,\mathrm{Gpc}$, plus D_ℓ^{fix}. Top: low ℓ (linear). Bottom: full spectrum. ΛCDM black, Planck grey, CoE one-sky realisation red in its $\pm 1\sigma$ band. $\chi^2_{2-2500} = 3{,}787$, Pearson $r^2 = 0.976$.

- The data fix only the local density gradient, $\rho'/\rho \approx -0.00106\,\mathrm{Gpc}^{-1}$; the density law away from the observed window is not determined.

- Shared harmonics $n = 1$–3 reduce the A residual (χ^2 25.54 to 11.46), and the leftover agrees in sign with 11 of 12 independent sky anomalies under an assignment fixed in advance.

- The CMB requires matter structure around the observer that the geometry alone does not supply; the best fit, $\chi^2_{2-2500} = 3{,}787$ and $r^2 = 0.976$, remains short of ΛCDM.

The inverse approach, solving for the density from the data, gives the model a working density law; the CMB and the origin of the half-scale mode are the open problems.

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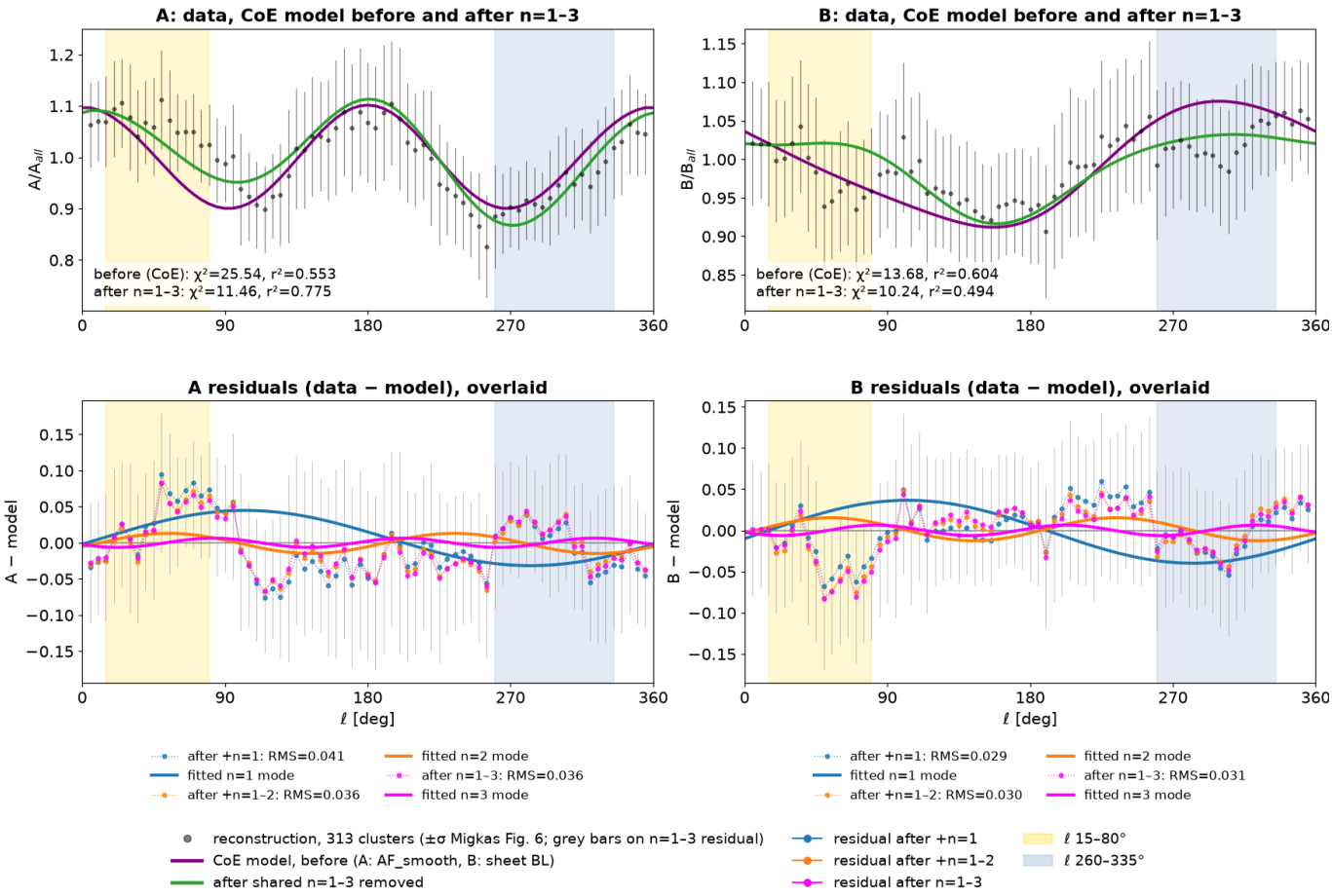


Figure 25: Harmonic decomposition, A (left) and B (right). Top: data with the CoE model before harmonics and after removing the shared $n = 1$–3 harmonics (χ^2, Pearson r^2 for both). Bottom: residuals (data − model) after $+n = 1$, $+n = 1$–2 and $n = 1$–3 overlaid, RMS in the legend.

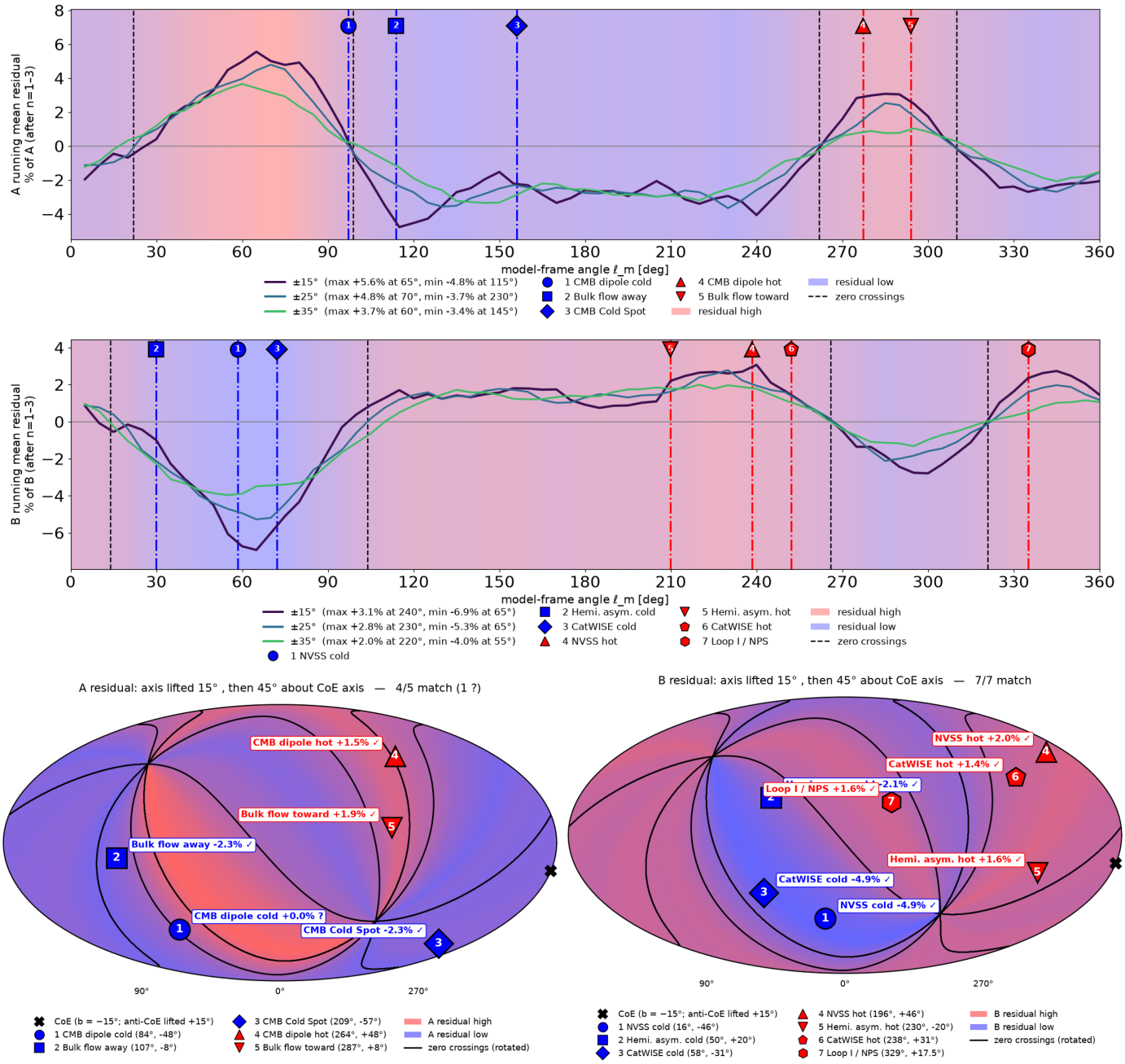


Figure 26: Top: A and B running-mean residuals vs model-frame angle (±15/25/35°). Bottom: residual maps (Galactic Mollweide, $\ell = 0$ at centre, increasing to the left) lifted 15° and rotated 45° about the CoE axis, with spots at published (ℓ, b); left A, right B.

Axis lifted 15°, then 45° counterclockwise about CoE axis — Score: A 4/5, B 7/7, total 11/12, 1 on the zero line (?)

Tied to	Spot	ℓ, b	Hot/cold	Needs	Residual at spot (%)	Match	Why A or B
A	CMB dipole cold	84, −48	cold	< 0	+0.02	?	Our motion → distance/velocity
A	Bulk flow away	107, −8	cold	< 0	−2.27	✓	Recession velocity → distance
A	CMB Cold Spot	209, −57	cold	< 0	−2.33	✓	Eridanus supervoid → distance
A	CMB dipole hot	264, +48	hot	> 0	+1.54	✓	Our motion → distance/velocity
A	Bulk flow toward	287, +8	hot	> 0	+1.90	✓	Recession velocity → distance
B	NVSS cold	16, −46	cold	< 0	−4.87	✓	Source density → sample
B	Hemi. asym. cold	50, +20	cold	< 0	−2.11	✓	Power direction → sample
B	CatWISE cold	58, −31	cold	< 0	−4.87	✓	Matter dipole → sample
B	NVSS hot	196, +46	hot	> 0	+2.03	✓	Source density → sample
B	Hemi. asym. hot	230, −20	hot	> 0	+1.62	✓	Power direction → sample
B	CatWISE hot	238, +31	hot	> 0	+1.39	✓	Matter dipole → sample
B	Loop I / NPS	329, +17.5	hot	> 0	+1.60	✓	Soft X-ray foreground → slope

Figure 27: Spot scores by sign of the tied residual. The qualitative assignment of each spot to A or B is given in the text below.